Companion Derivations to Solvency Field Theory (V10)

Combined reading volume — April 2026

This stitched volume combines the preserved companion chapters in reading order: C0–C1, C3–4, C5, C6, C7, C8, C9, C10, C11, C12, and Appendices A–D.

It is assembled from the checkpoint PDFs so the original modular files remain unchanged.

Included source volumes

- Part I: C5, C8, C10
- Part II: C0, C1, C3–4, C6, C7, C9, Appendices
- Standalone C11
- Standalone C12

Simple stitched checkpoint edition.

1. C0. How to Use This Companion

Purpose: This document is the mathematical audit trail for *Solvency Field Theory: An Information-Theoretic Foundation for Cosmology and Gravity* (V10). The master paper states the framework, its quantitative claims, and the logical skeleton of the derivations. This companion carries the full mathematical derivations, numerical methodology notes, status labels, and audit trails for the load-bearing Sections 5–12. It is not a second master paper and is not intended to be read linearly by a first-time reader. It is intended to be consulted alongside the master whenever a claim needs to be checked in full detail.

1.1 C0.1 Scope

The companion covers the parts of V10 where a technically skeptical reader would reasonably demand explicit derivations, numerical pipelines, or theorem-status clarification. The primary chapters are:

- **C5** – full derivation of the modified Friedmann equation,
- **C6** – weak-field screen-energy and SPARC audit,
- **C8** – holographic count equation and constrained minisuperspace action,
- **C10** – cosmological sector, four-state ladder, TNG300 methodology, historical pipeline, corrected $w_{\mathrm{DE}}(z)$ chain, and audit note,
- **C11** – early-universe shatter, projection program, transfer-budget extraction, and candidate CMB splice,
- **C12** – strong-field closure, Schwarzschild recovery, and TOV recovery.

Supporting chapters provide notation, status mapping, foundational clarifications, and structural summaries. Introductory framing, broad motivation, prediction summaries, and conclusion material remain in the master paper and are not duplicated here except where needed for precision or context.

1.2 C0.2 Mapping to the Master Paper

This companion mirrors the master selectively rather than line-by-line.

Master section	Companion material	Role
Sections 1–2	none	framing stays in master
Sections 3–4	C3–4	technical scaffold only
Section 5	C5	full algebraic derivation
Section 6	C6	full weak-field derivation + SPARC methods
Section 7	C7	structural-identity map
Section 8	C8	full count-equation and action derivation
Section 9	C9	forced-transition scaffold
Section 10	C10	full cosmology derivations + audit
Section 11	C11	early-universe candidate derivations
Section 12	C12	strong-field uniqueness / closure
Sections 13–15	appendices / status ledger only	no full derivation chapters

This is exactly the split implied by V10 itself: the master carries the conceptual argument, while the companion carries the full derivations for the load-bearing mathematical sectors.

1.3 C0.3 Epistemic Status Labels

Every companion chapter uses the following status labels.

- **Derived** – the result follows within the stated scaffold and the derivation is carried explicitly here.
- **Empirically confirmed** – the derivation is complete and the associated quantitative test already passed in the framework's current evidence base.
- **Candidate splice** – a numerically viable and physically motivated closure exists, but the unbroken symbolic theorem is not yet written.
- **Under audit** – the formal chain has been corrected or sharpened, but a full numerical or methodological rerun is still pending.
- **Open theorem** – the result is not yet proved and is explicitly named as an open target in the master paper.

These labels are not decorative. V10 repeatedly distinguishes fully derived results from candidate and audit-status results, especially in Sections 7, 10, 11, and 14, and this companion preserves that distinction rather than smoothing it over.

1.4 C0.4 How to Read This Document

A first-time reader should read the master paper first.

A skeptical technical reader should use the companion in one of three ways:

1. **Claim audit** – jump directly from a master-paper claim to its matching companion chapter.
2. **Sector audit** – read a full domain chapter such as C6, C10, or C12.
3. **Status audit** – consult the status ledger and appendices to see which parts are derived, confirmed, candidate, or still open.

The recommended first-audit order is:

$$C5 \to C8 \to C10 \to C12 \to C6 \to C11.$$

That order follows the framework's main logical load-bearing chain: cosmological backbone, variational scaffold, late-universe sector, strong-field closure, weak-field confirmation, then early-universe candidate closure.

2. C1. Notation, Conventions, and Constants

Purpose: Fix notation once so later chapters can stay focused on derivations rather than bookkeeping.

2.1 C1.1 Constants

The following constants are used throughout:

$c =$ speed of light, $\quad G =$ Newtonian gravitational constant, $\quad h =$ linear Planck constant, $\quad \hbar = \dfrac{h}{2\pi}, \quad k$

The Planck area is

$$\ell_P^2 = \frac{\hbar G}{c^3} = \frac{hG}{2\pi c^3}.$$

Numerically,

$$\ell_P \approx 1.616 \times 10^{-35}\ \mathrm{m}, \qquad \ell_P^2 \approx 2.61 \times 10^{-70}\ \mathrm{m}^2.$$

The exponent 10^{-70} is the correct one; earlier appearances of $10^{-60}\,\mathrm{m}^2$ in development drafts were typographical errors and are not to be used. This companion adopts the corrected value consistently.

2.2 C1.2 Linear vs angular phase conventions

A major structural distinction in SFT is the use of **linear-cycle counting** in the Compton Maintenance Postulate.

The Compton frequency in linear-cycle form is

$$\nu_C = \frac{mc^2}{h},$$

while the angular form is

$$\omega_C = \frac{mc^2}{\hbar} = 2\pi\nu_C.$$

SFT's load-bearing convention is:

one irreversible bit = one complete phase cycle, not one radian of phase advance.

This means bulk ticking rates are written using h, while Planck-scale and horizon formulas are often written using $\hbar$. The conversion

$$h = 2\pi\hbar$$

therefore matters physically and not just notationally. This is one of the places where the recurring 2π factors enter the theory's structural identity.

2.3 C1.3 Horizon conventions

In the FLRW minisuperspace sector, the relevant boundary is the Hubble / apparent horizon:

$$R_H = \frac{c}{H}.$$

Its associated area and volume are

$$A_H = 4\pi R_H^2 = 4\pi \left(\frac{c}{H}\right)^2,$$

$$V_H = \frac{4\pi}{3} R_H^3 = \frac{4\pi}{3} \left(\frac{c}{H}\right)^3.$$

In the flat-FLRW setting used in the master and companion background derivations, the Hubble sphere is treated as the relevant apparent-horizon bookkeeping boundary for one-Hubble-time settlement of the information ledger.

2.4 C1.4 Density and mass conventions

The mean cosmic density is ρ. When needed, the enclosed Hubble-volume mass is

$$M = \rho V_H.$$

In the cosmological-sector derivations, ρ_m denotes pressureless matter density and $\rho_{\rm DE}$ denotes the effective dark-energy residual reconstructed from the expansion history:

$$\rho_{\rm DE}(z) \equiv \frac{3H(z)^2}{8\pi G} - \rho_{m,0}(1+z)^3.$$

In the strong-field sector, $m(r)$ denotes the enclosed Misner–Sharp-type mass function in static spherical symmetry.

2.5 C1.5 FLRW conventions

The background cosmology uses a flat FLRW metric with scale factor $a(t)$ and Hubble parameter

$$H = \frac{\dot{a}}{a}.$$

The logarithmic e-fold variable is

$$N = \ln a.$$

In the corrected cosmological reconstruction chain, the effective total equation of state is

$$w_{\rm eff}(z) = -1 + \frac{1+z}{3}\frac{d\ln H^2}{dz},$$

while the reconstructed dark-energy residual is

$$w_{\rm DE}(z) = -1 + \frac{1+z}{3\,\Omega_{\rm DE}(z)}\frac{d\Omega_{\rm DE}}{dz},$$

with

$$\Omega_{\rm DE}(z) = E(z)^2 - \Omega_{m0}(1+z)^3, \qquad E(z)^2 = \frac{H(z)^2}{H_0^2}.$$

These formulas are used only where explicitly stated. The companion will always distinguish between $w_{\rm eff}$ and $w_{\rm DE}$, since V10 now treats that distinction as load-bearing in the cosmological audit.

2.6 C1.6 Strong-field conventions

For static spherical symmetry, two gauges appear.

2.6.1 Exponential / reciprocal gauge

$$ds^2 = -e^{-2U(r)}c^2dt^2 + e^{+2U(r)}dr^2 + r^2d\Omega^2.$$

2.6.2 Standard mass-function gauge

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \left(1 - \frac{2Gm(r)}{c^2r}\right)^{-1}dr^2 + r^2d\Omega^2.$$

These are two parameterizations of the same strong-field closure sector, and the companion uses whichever is cleaner for the derivation being discussed.

2.7 C1.7 Status conventions inside derivations

Whenever a displayed result appears, it should be understood in one of three ways:

1. **Exact result in current scaffold**
2. **Leading closure / leading candidate in current scaffold**
3. **Candidate splice under active theorem development**

The companion explicitly marks which is which. This prevents exploratory results from being accidentally read as completed theorems.

3. C3–4. Foundational Scaffold and the Compton Maintenance Note

Master sections mirrored: Sections 3 and 4

Claim being derived: The master equation

$$H = \frac{\dot{S}}{I}$$

is the native thermodynamic identity of the SFT scaffold, and the Compton Maintenance Postulate is the physical mechanism that makes $\dot{S}$ calculable.

Inputs / assumptions:

1. Landauer principle
2. Connes–Rovelli thermal time hypothesis
3. Holographic bound / horizon information capacity
4. Expansion as geometric accommodation of irreversible information production
5. Continuous Compton-frequency phase evolution of massive particles

Result: The foundational layer of SFT is a scaffold consisting of:

- four axioms,
- one master identity,
- one mechanism postulate.

This chapter clarifies what is assumption, what is consequence, and what remains interpretation.

Status: Foundational scaffold / not a long-form derivation chapter.

3.1 C3–4.1 The four axioms in technical form

The four axioms are:

1. **Information has thermodynamic cost.** Irreversible information processing is constrained by Landauer's principle.
2. **Time is the entropy clock.** Macroscopic time tracks irreversible state change rather than existing as a detached external background.
3. **The holographic bound is absolute.** Information capacity scales with boundary area and is capped by horizon geometry.
4. **Expansion is the geometric cost of maintenance.** If irreversible processing continues while the boundary capacity is finite, cosmological geometry must grow to accommodate that processing.

V10 is explicit that SFT's novelty is not inventing these ingredients individually but committing to all four simultaneously and following the joint consequences.

3.2 C3–4.2 Why $H = \dot{S}/I$ is the native identity of the scaffold

The master equation

$$H = \frac{\dot{S}}{I}$$

should not be read as an arbitrary dynamical postulate inserted into the theory. Within the stated scaffold, it is the natural fractional-rate statement:

- $\dot{S}$ is the rate of bulk irreversible processing,
- I is the current boundary information capacity,
- $\dot{S}/I$ is therefore the fractional rate at which the ledger is being filled,
- and H is the fractional rate of cosmological expansion.

So the equation identifies these two fractional rates. That is why V10 now calls it the **native thermodynamic identity** of the framework rather than a brute dynamical law.

What this does **not** mean is that the equation is derivable from pure mathematics without physical assumptions. It is native to the scaffold, not assumption-free in the abstract.

3.3 C3–4.3 The Compton Maintenance Postulate

The Compton Maintenance Postulate supplies the physical content of $\dot{S}$.

For a particle of rest mass m, the linear Compton frequency is

$$\nu_C = \frac{mc^2}{h}.$$

The postulate states that:

- a massive particle maintains its quantum existence through continuous phase evolution at the Compton frequency,
- each complete phase cycle counts as one irreversible bit,
- and each such bit requires one Planck area of cosmological boundary accommodation.

This is the move that turns the axiomatic scaffold into something calculable. Without it, $\dot{S}$ remains formal. With it, the total bulk maintenance rate becomes

$$\dot{S} = \frac{Mc^2}{h}$$

for total mass-energy M inside the Hubble volume.

3.4 C3–4.4 One cycle versus one radian

This is one of the most important hidden choices in the whole framework.

The postulate chooses:

$$\text{one bit} = \text{one complete phase cycle.}$$

That means the counting convention uses h, not $\hbar$.

If instead one chose one radian of phase advance as the fundamental irreversible unit, then the rate would differ by a factor of 2π, and the major quantitative consequences of the theory – the modified Friedmann ratio, the galactic acceleration threshold, and the structural $4\pi^2$ identity – would all shift accordingly.

This is why the h-versus-$\hbar$ choice is not a stylistic notation issue but a load-bearing physical convention in SFT.

3.5 C3–4.5 One Planck area per bit

The second half of the mechanism is the geometric cost:

$$\text{one irreversible bit} \longrightarrow \text{one Planck area of accommodation.}$$

This is consistent with the holographic bound in the sense that a horizon of area A has information capacity

$$I = \frac{A}{4\ell_P^2}.$$

The Compton Maintenance Postulate promotes this from a static capacity statement to a process statement: bulk processing continuously consumes boundary accommodation.

This is why the master equation and the count equation are both possible:

- the master equation compares **rates**,
- the count equation compares **one-Hubble-time accumulated counts**.

3.6 C3–4.6 What this foundational layer does and does not claim

This foundational chapter makes the following claims:

- the SFT scaffold is built from four explicit axioms,
- the master equation is the native fractional-rate identity inside that scaffold,
- the Compton Maintenance Postulate provides the mechanism making $\dot{S}$ calculable.

It does **not** claim:

- that the axioms are derivable from one another,
- that the master equation is a theorem independent of the scaffold,
- or that the Compton Maintenance Postulate has already been reduced to a deeper microscopic theory.

Those are stronger claims than the framework currently needs to make.

3.7 C3–4.7 Why this chapter is short

The master paper uses Sections 3 and 4 to tell the reader what kind of physical object the universe is in SFT. That is fundamentally a **framework declaration**, not a long algebraic derivation. The role of the companion is therefore only to clarify the exact technical commitments and prevent later chapters from silently changing the interpretation.

So this chapter is intentionally short:

- enough to define the scaffold,
- not long enough to become a second introduction.

4. C6. Weak-Field Sector and SPARC Audit

Master section mirrored: Section 6

Claim being derived: The weak-field galactic sector of SFT is controlled by the acceleration ratio

$$x = \frac{g_{\mathrm{bar}}}{a_{\mathrm{SFT}}}, \qquad a_{\mathrm{SFT}} = \frac{cH_0}{2\pi},$$

with an internally motivated response family

$$g_{\mathrm{eff}}^2 = g_{\mathrm{bar}}^2 + \lambda\, a_{\mathrm{SFT}}\, g_{\mathrm{bar}}, \qquad \nu(x) = \sqrt{1 + \frac{\lambda}{x}},$$

1. C5. Full Derivation of the Modified Friedmann Equation

Master section mirrored: Section 5

Claim being derived: The SFT master equation together with the Compton Maintenance Postulate yields

$$H^2 = \frac{2G\rho}{3\pi},$$

and therefore

$$\frac{H^2_{\mathrm{SFT}}}{H^2_{\mathrm{Friedmann}}} = \frac{1}{4\pi^2}.$$

Inputs / assumptions:

1. Master equation
$$H = \frac{\dot{S}}{I}$$
2. Compton maintenance rate
$$\dot{S} = \frac{Mc^2}{h}$$
3. Hubble-horizon information capacity
$$I = \frac{A_H}{4\ell_P^2}$$
4. Flat FLRW Hubble horizon
$$R_H = \frac{c}{H}, \qquad A_H = 4\pi R_H^2 = 4\pi \left(\frac{c}{H}\right)^2$$
5. Hubble-volume mass content
$$M = \rho V_H, \qquad V_H = \frac{4\pi}{3}\left(\frac{c}{H}\right)^3$$
6. Planck area
$$\ell_P^2 = \frac{\hbar G}{c^3} = \frac{hG}{2\pi c^3}$$

Result: The modified Friedmann equation is

$$H^2 = \frac{2G\rho}{3\pi},$$

and relative to standard flat-dust Friedmann,

$$H^2_{\mathrm{std}} = \frac{8\pi G\rho}{3},$$

the ratio is

$$\frac{H^2_{\mathrm{SFT}}}{H^2_{\mathrm{std}}} = \frac{1}{4\pi^2}.$$

Status: Derived.

1.1 C5.1 Starting point: the master equation

Within the SFT scaffold, the Hubble parameter is identified with the fractional rate at which bulk irreversible processing fills the available holographic boundary capacity:

$$H = \frac{\dot{S}}{I}.$$

To turn this identity into a cosmological equation, both $\dot{S}$ and I must be written in terms of the matter content and the Hubble-scale geometry.

1.2 C5.2 Bulk maintenance rate

The Compton Maintenance Postulate states that each massive particle contributes one irreversible bit per complete Compton cycle. Summed over all mass-energy M within the Hubble volume, the total maintenance rate is

$$\dot{S} = \frac{Mc^2}{h}.$$

This is the total number of complete phase cycles per unit time, counted in the linear-cycle convention using h, not the angular convention using $\hbar$.

1.3 C5.3 Boundary information capacity

The Hubble horizon has radius

$$R_H = \frac{c}{H},$$

so its area is

$$A_H = 4\pi \left(\frac{c}{H}\right)^2.$$

Applying the Bekenstein-Hawking counting rule, the information capacity is

$$I = \frac{A_H}{4\ell_P^2} = \frac{4\pi(c/H)^2}{4\ell_P^2} = \frac{\pi c^2}{\ell_P^2 H^2}.$$

1.4 C5.4 Substitute into the master equation

Insert the bulk rate and boundary capacity into

$$H = \frac{\dot{S}}{I}.$$

This gives

$$H = \frac{Mc^2/h}{\pi c^2/(\ell_P^2 H^2)} = \frac{Mc^2}{h} \cdot \frac{\ell_P^2 H^2}{\pi c^2}.$$

Canceling c^2,

$$H = \frac{M\ell_P^2 H^2}{\pi h}.$$

At this stage the equation is still written in terms of the total enclosed mass M.

1.5 C5.5 Replace M by the Hubble-volume mass

For a homogeneous mean density ρ,

$$M = \rho V_H = \rho \frac{4\pi}{3} \left(\frac{c}{H} \right)^3 .$$

Substitute this into the previous equation:

$$H = \frac{\rho \frac{4\pi}{3} (c/H)^3 \ell_P^2 H^2}{\pi h}.$$

Cancel one factor of π:

$$H = \frac{4}{3} \rho \frac{c^3}{H^3} \frac{\ell_P^2 H^2}{h}.$$

Simplify the Hubble powers:

$$H = \frac{4}{3} \rho \, \frac{c^3 \ell_P^2}{h} \, \frac{1}{H}.$$

Multiply both sides by H:

$$H^2 = \frac{4}{3} \rho \, \frac{c^3 \ell_P^2}{h}.$$

Everything now reduces to the explicit form of the Planck area.

1.6 C5.6 The h-versus-$\hbar$ step

The Planck area is conventionally defined as

$$\ell_P^2 = \frac{\hbar G}{c^3}.$$

To keep the counting convention consistent with the Compton maintenance rate, rewrite $\hbar$ in terms of h:

$$\hbar = \frac{h}{2\pi}.$$

Therefore

$$\ell_P^2 = \frac{hG}{2\pi c^3}.$$

Substitute this into the previous expression:

$$H^2 = \frac{4}{3} \rho \, \frac{c^3}{h} \cdot \frac{hG}{2\pi c^3}.$$

Now the cancellations are explicit:

- h cancels
- c^3 cancels

leaving

$$H^2 = \frac{4}{3}\rho\frac{G}{2\pi}.$$

Thus

$$H^2 = \frac{2G\rho}{3\pi}.$$

This is the SFT modified Friedmann equation.

1.7 C5.7 Comparison to standard Friedmann

For flat FLRW with pressureless dust, standard general relativity gives

$$H_{\text{std}}^2 = \frac{8\pi G\rho}{3}.$$

The SFT-to-standard ratio is therefore

$$\frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{\frac{2G\rho}{3\pi}}{\frac{8\pi G\rho}{3}}.$$

Cancel G, ρ, and the factor of 3:

$$\frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{2}{8\pi^2} = \frac{1}{4\pi^2}.$$

So the modified Friedmann law differs from the standard one by the exact content-independent factor

$$\boxed{\frac{1}{4\pi^2}}.$$

1.8 C5.8 Interpretation of the two 2π factors

The ratio

$$\frac{1}{4\pi^2}$$

is the first appearance of the structural identity developed later in the master paper.

At the algebraic level, one factor of 2π enters because the bulk processing rate is written in the linear-cycle convention using h, while the Planck area is conventionally defined using $\hbar$. The second factor of 2π appears only when the SFT result is compared to the standard Friedmann equation, whose normalization inherits the Einstein-side $8\pi G$ bulk coupling rather than the Bekenstein-Hawking boundary counting convention. The full structural interpretation belongs to Companion Chapter C7; here the important point is simply that the ratio is exact and content-independent.

1.9 C5.9 Dimensional check

The final equation is

$$H^2 = \frac{2G\rho}{3\pi}.$$

Check dimensions:

- H^2 has units of T^{-2}
- G has units $L^3M^{-1}T^{-2}$
- ρ has units ML^{-3}

Therefore

$$G\rho \sim T^{-2},$$

so the dimensions are correct.

1.10 C5.10 Matter-only amplitude and the structural gap

The matter-only SFT equation does not by itself reproduce the observed present-day expansion rate. If one writes

$$H^2_{\mathrm{SFT}} = \frac{2G\rho_m}{3\pi},$$

and evaluates it for the present matter fraction $\Omega_m \approx 0.31$, then the resulting matter-only amplitude is below the observed H_0, leaving the structural gap discussed in the master paper.

That gap is not repaired inside this chapter. This chapter establishes only the exact matter-only backbone:

$$H^2 = \frac{2G\rho}{3\pi},$$

together with its exact ratio

$$\frac{1}{4\pi^2}$$

to the standard Friedmann form.

The cosmological-sector question is then: how do the late-time maintenance-regime dynamics modify the effective expansion history away from the matter-only backbone? That is the subject of Companion Chapter C10.

1.11 C5.11 What this chapter has established

Starting from the master equation

$$H = \frac{\dot{S}}{I},$$

and using the Compton Maintenance Postulate together with the Hubble-horizon capacity, the SFT cosmological background law is

$$\boxed{H^2 = \frac{2G\rho}{3\pi}}.$$

Relative to standard flat-dust Friedmann,

$$\boxed{\frac{H_{\mathrm{SFT}}^2}{H_{\mathrm{std}}^2} = \frac{1}{4\pi^2}}.$$

No fitting constants enter the derivation. The entire result follows from the bulk maintenance rate, the boundary counting rule, the Hubble-volume mass content, and the h-versus-$\hbar$ conversion built into the Compton-cycle convention.

and with benchmark coefficients

$$\lambda_{\rm sph} = \frac{4}{3}, \qquad \lambda_{\rm disk} = \frac{3\pi}{8},$$

where the spherical value is the derived isotropic limit and the disk value is the leading current axial-projection candidate. The SPARC data then test both the acceleration scale and the morphology dependence.

Inputs / assumptions:

1. The weak-field control variable is an acceleration ratio, not the strong-field compactness variable χ.
2. The natural SFT weak-field scale is
$$a_{\rm SFT} = \frac{cH_0}{2\pi}.$$
3. The galactic response is described by a solvency-family interpolation law.
4. The observable coefficient λ depends on the projection geometry of the screen onto the source.
5. The isotropic spherical-screen limit is exact within the current construction; the disk-sector value is a sharply motivated candidate, not yet a closed theorem.

Result: The weak-field sector predicts a universal scale $a_{\rm SFT}$ and a morphology-sensitive coefficient λ. Against 137 quality-filtered SPARC galaxies, the derived scale

$$a_{\rm SFT} = \frac{cH_0}{2\pi} \approx 1.05 \times 10^{-10}\ \mathrm{m\,s^{-2}}$$

reduces the mean relative RMSE from 0.458 to 0.167, outperforming cH_0 and $cH_0/6$. The morphology dependence is confirmed at

$$p = 0.0016,$$

with disky systems clustering near $3\pi/8$ and bulge-rich systems near $4/3$.

Status: Weak-field scale derived and empirically confirmed; spherical λ derived; disk λ a leading candidate; morphology dependence empirically confirmed; full morphology-to-λ mapping remains open.

4.1 C6.1 Why the weak-field sector needs its own control variable

The strong-field compactness variable

$$\chi(r) = \frac{GM(r)}{2c^2 r}$$

is natural for compact objects, but for ordinary disk galaxies it is far too small to generate the observed rotation-curve departures from baryonic Newtonian predictions. Earlier weak-field attempts built directly on χ were therefore effectively locked to the baryons-only limit. The weak-field sector only became viable when the control variable was replaced by the acceleration ratio

$$x = \frac{g_{\rm bar}}{a_{\rm SFT}}.$$

This is the first conceptual closure of the galactic sector: galaxies probe a low-acceleration response law rather than a compactness perturbation theory.

4.2 C6.2 Derived weak-field scale

The modified Friedmann ratio from C5 is

$$\frac{H_{\mathrm{SFT}}^2}{H_{\mathrm{std}}^2} = \frac{1}{4\pi^2}.$$

Taking the square root exposes one factor of 2π, and multiplying by cH_0 yields the natural weak-field acceleration scale

$$\boxed{a_{\mathrm{SFT}} = \frac{cH_0}{2\pi}.}$$

Numerically,

$$a_{\mathrm{SFT}} \approx 1.05 \times 10^{-10}\ \mathrm{m\,s^{-2}}.$$

This is not a fitted MOND-like scale imported from phenomenology; it is the weak-field manifestation of the same phase-convention factor already present in the cosmological backbone. V10 therefore treats the SPARC sector as the direct empirical test of whether the complete-cycle convention is physically real.

4.3 C6.3 Weak-field response family

The current weak-field law is written as

$$g_{\mathrm{pred}}(r) = g_{\mathrm{bar}}(r)\,\nu\!\left(\frac{g_{\mathrm{bar}}(r)}{a_{\mathrm{SFT}}}\right),$$

with the solvency-family form

$$g_{\mathrm{eff}}^2 = g_{\mathrm{bar}}^2 + \lambda\, a_{\mathrm{SFT}}\, g_{\mathrm{bar}},$$

equivalently

$$\boxed{\nu(x) = \sqrt{1 + \frac{\lambda}{x}}, \qquad x = \frac{g_{\mathrm{bar}}}{a_{\mathrm{SFT}}}.}$$

This family has the correct asymptotics:

- for $x \gg 1$, $\nu(x) \to 1$, so the Newtonian limit is recovered;
- for $x \ll 1$, $g_{\mathrm{eff}} \propto \sqrt{g_{\mathrm{bar}} a_{\mathrm{SFT}}}$, giving the standard deep weak-field scaling.

The current companion treats this family as the internally motivated response law, with the remaining question being not whether the family exists, but what fixes λ.

4.4 C6.4 Screen-energy interpretation

The master paper summarizes the weak-field derivation in three-sentence form: the local information-maintenance cost is governed by a screen-energy functional; the effective acceleration is the gradient of that screen-energy functional; and the self-consistency of the screen with the baryonic mass distribution closes the response law into the solvency-family interpolation. The companion does not need to invent a more elaborate story than that. The essential point is that $\nu(x)$ is not introduced as a purely empirical MOND ansatz. It is motivated by how much of the local maintenance burden is absorbed by pre-existing curvature versus transmitted outward to the cosmological boundary.

4.5 C6.5 Spherical-screen limit

The first exact benchmark for λ is the isotropic spherical-screen limit. In that limit, the observable circular-support response is weighted by the tangential projector $\sin^2\theta$ over the sphere, so

$$\lambda_{\rm sph} = 2\left\langle \sin^2\theta \right\rangle_{\mathbb{S}^2}.$$

Using the normalized spherical average,

$$\langle f(\theta,\phi)\rangle_{\mathbb{S}^2} = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} f(\theta,\phi)\sin\theta\, d\theta\, d\phi,$$

we get

$$\left\langle \sin^2\theta \right\rangle_{\mathbb{S}^2} = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} \sin^2\theta\, \sin\theta\, d\theta\, d\phi = \frac{2}{3}.$$

Therefore

$$\boxed{\lambda_{\rm sph} = 2\cdot\frac{2}{3} = \frac{4}{3}.}$$

This is the robust isotropic spherical-screen limit of the current weak-field construction. It is the more conservative derived benchmark carried by V10.

4.6 C6.6 Why raw disk field geometry fails

A natural next attempt is to derive the disk coefficient directly from the raw Newtonian field geometry of an ideal thin disk projected onto a spherical screen. Earlier internal tests did exactly that and found coefficients in the wrong direction, around 1.60–1.73, rather than the observed ~ 1.2 range. This is an important negative result: the suppression needed to move from the spherical value toward the disky value is not explained by raw field-strength geometry alone. That failure is what motivates the axial-projection candidate rather than a direct field-magnitude derivation.

4.7 C6.7 Disk-sector axial-projection candidate

The strongest current disk-sector derivation candidate is symmetry-based. Let $\hat{n}$ be the disk rotation axis, and let $\xi_{\rm ax}$ denote the azimuthal Killing field around that axis. At screen direction $\hat{r}$, the normalized participation field is

$$w(\hat{r};\hat{n}) = \frac{|\xi_{\rm ax}|}{r} = \sin\theta,$$

where θ is the polar angle relative to the rotation axis. The disk-sector coefficient is then

$$\lambda_{\rm disk} = 2\left\langle w(\theta)\sin^2\theta \right\rangle_{\mathbb{S}^2} = 2\left\langle \sin\theta\, \sin^2\theta \right\rangle_{\mathbb{S}^2}.$$

Because the spherical measure contributes another factor of $\sin\theta$, this becomes

$$\lambda_{\rm disk} = 2\cdot\frac{1}{2}\int_0^{\pi} \sin^4\theta\, d\theta.$$

Using

$$\int_0^{\pi} \sin^4\theta\, d\theta = \frac{3\pi}{8},$$

we obtain

$$\boxed{\lambda_{\text{disk}} = \frac{3\pi}{8} \approx 1.1781.}$$

This is not yet claimed as a theorem in the master. It is the strongest current disk-sector candidate: symmetry-motivated, numerically right, and much cleaner than the raw disk-field attempt.

4.8 C6.8 SPARC sample and prediction pipeline

The observational test uses the SPARC database of observed rotation curves and baryonic mass profiles derived from 3.6 μm Spitzer photometry. Starting from 175 nearby disk galaxies, standard quality filters are applied:

- inclination $> 30°$,
- distance < 100 Mpc,
- sufficient radial coverage.

After filtering, 137 galaxies remain in the test sample. For each galaxy, the predicted rotation curve is computed using only the observed baryonic mass profile and a single universal acceleration scale:

$$g_{\text{pred}}(r) = g_{\text{bar}}(r)\, \nu\!\left(\frac{g_{\text{bar}}(r)}{a_{\text{SFT}}}\right).$$

No per-galaxy halo parameters or fit coefficients are introduced in this comparison.

4.9 C6.9 SPARC comparison table

The load-bearing direct comparison uses four acceleration scales with the same interpolation-function form and the same quality filters:

- baryons only: relRMSE $= 0.458$
- SFT:
$$a = \frac{cH_0}{2\pi}$$
relRMSE $= 0.167$
- alternative:
$$a = cH_0$$
relRMSE $= 0.283$
- alternative:
$$a = \frac{cH_0}{6}$$
relRMSE $= 0.174$

So the SFT-derived scale reduces the mean relative RMSE by 63% relative to baryons-only, outperforms the literal cH_0 scale, and slightly outperforms the nearby $cH_0/6$ alternative as well. This is exactly the point the master emphasizes: the comparison exists, and the data select the derived 2π factor before the broader theoretical story is even invoked.

4.10 C6.10 What the comparison actually proves

The comparison table supports two distinct statements.

First, it says that a scale of order cH_0 is physically relevant to the weak-field galactic regime. That is the broad closure of the weak-field sector.

Second, it says that the specific factor of 2π predicted by the Compton Maintenance Postulate is not just rhetorical decoration. The scale $cH_0/(2\pi)$ beats the literal cH_0 alternative directly. The $cH_0/6$ alternative comes close because 6 numerically approximates 2π, which is exactly why the master is careful not to oversell the discrimination as stronger than the present sample warrants. But the preference is still in the correct direction.

4.11 C6.11 Morphology dependence

The next-level test is whether λ is universal. The current framework says it is not: λ depends on the projection geometry of the local screen onto the baryonic source. In the present state of the theory, the two load-bearing reference values are:

$$\boxed{\lambda_{\text{sph}} = \frac{4}{3}, \qquad \lambda_{\text{disk}} = \frac{3\pi}{8}.}$$

The master paper is explicit about the status split:

- the spherical value is the conservative derived reference,
- the disk value is the leading axial-projection candidate,
- and the full morphology-to-λ mapping remains an open derivation problem for the companion/theorem program.

4.12 C6.12 Morphology statistics

The empirical morphology signal is real. Using bulge fraction as the morphological covariate and controlling for additional structural and photometric variables through a partial-correlation test, the λ coefficient correlates with bulge fraction at

$$p = 0.0016.$$

Earlier internal summary notes give the same qualitative picture numerically:

- low-bulge / disky galaxies cluster near

  $$\lambda \approx 1.14,$$

  close to $3\pi/8 \approx 1.178$,
- bulge-rich galaxies cluster near

  $$\lambda \approx 1.42,$$

  close to $4/3 \approx 1.333$,

- mixed galaxies lie between them.

So the correct present statement is:

- morphology dependence is empirically confirmed,
- the endpoint benchmarks are structurally meaningful,
- the continuous mapping between real morphology and effective screen geometry is still open.

4.13 C6.13 What does not explain the coefficient

The current weak-field notes also report a useful negative result: inclination by itself is too weak to explain the observed spread in fitted λ. The data show some dependence on internal structure – bulge fraction, gas fraction, compactness, morphology – but not enough support for the idea that λ is merely a trivial viewing-angle artifact. That matters because it keeps the problem physical: the coefficient remains fundamentally a response-law / screen-geometry question rather than collapsing into a pure orientation nuisance parameter.

4.14 C6.14 Link to the cosmological projection program

V10 explicitly says that the projection integrals responsible for the galactic λ benchmarks are the same projection family that later produces

$$s_{\text{wall}} = \frac{1}{3}, \qquad s_{\text{filament}} = \frac{2}{3}$$

in the cosmological ladder. This is the reason C6 and C10 belong in the same companion volume: they are not isolated observational sectors but two scales of one projection-integral mechanism. In the present framework, this unification is stated structurally and kept as an open theorem target rather than a finished proof.

4.15 C6.15 What this chapter has established

The weak-field companion chapter establishes the following:

1. The correct weak-field control variable is the acceleration ratio

$$x = \frac{g_{\text{bar}}}{a_{\text{SFT}}},$$

   not the strong-field compactness burden χ.

2. The natural SFT acceleration scale is

$$\boxed{a_{\text{SFT}} = \frac{cH_0}{2\pi},}$$

   the square-root weak-field manifestation of the Friedmann-sector $1/(4\pi^2)$ identity.

3. The response family is

$$g_{\text{eff}}^2 = g_{\text{bar}}^2 + \lambda a_{\text{SFT}} g_{\text{bar}}, \qquad \nu(x) = \sqrt{1 + \lambda/x}.$$

4. The isotropic spherical benchmark is derived:

$$\boxed{\lambda_{\text{sph}} = \frac{4}{3}.}$$

5. The leading current disk-sector axial-projection candidate is

$$\boxed{\lambda_{\text{disk}} = \frac{3\pi}{8}.}$$

6. The SPARC comparison on 137 quality-filtered galaxies gives relRMSE values

$$0.458,\ 0.167,\ 0.283,\ 0.174$$

for baryons-only, $cH_0/(2\pi)$, cH_0, and $cH_0/6$ respectively, yielding a 63% error reduction for the derived scale.

7. Morphology dependence is empirically confirmed at

$$p = 0.0016,$$

with disky systems near $3\pi/8$ and bulge-rich systems near $4/3$, while the full morphology-to-screen-geometry derivation remains open.

So the weak-field sector is now in its strongest honest form:

- the scale is derived,
- the spherical coefficient is derived,
- the disk coefficient is a sharply motivated candidate,
- the morphology dependence is real,
- and the remaining open step is the first-principles continuous mapping from observed galaxy structure to effective screen geometry.

5. C7. Dependency Map for the Five Appearances of $4\pi^2$

Master section mirrored: Section 7

Claim being derived: The recurring factor

$$4\pi^2 = (2\pi)(2\pi)$$

appears in five distinct physical contexts in SFT. Four appearances are derived or structurally implied within the current scaffold; the fifth is a candidate splice. This chapter maps where each appearance comes from, what physical role it plays, and what its current epistemic status is.

Inputs / assumptions:

1. Linear-cycle versus angular-phase counting distinction
2. Boundary-versus-bulk normalization distinction
3. Modified Friedmann derivation
4. Weak-field acceleration-threshold derivation
5. Holographic count comparison
6. Cosmological structural gap and early-universe candidate splice

Result: The five appearances are not independent curiosities. They form a dependency spine:

- Appearance 1 is the foundational ratio,
- Appearance 2 is its square-root weak-field manifestation,
- Appearance 3 is the same counting mismatch viewed as a boundary/bulk capacity comparison,
- Appearance 4 is the late-universe structural gap built on the same backbone,
- Appearance 5 is the candidate CMB transfer splice.

Status: Four appearances derived or structurally anchored; one candidate splice.

5.1 C7.1 Why this chapter is a map and not a full derivation

The master paper already gives the conceptual unification claim: the same numerical structure appears in five distinct places, and in each case it decomposes into two independent 2π factors – one associated with the linear-versus-angular phase convention, one associated with boundary-versus-bulk normalization. The companion should not duplicate all the later derivations here. Its job is to show **where each appearance lives**, **which chapter derives it**, and **what status it has**.

So this chapter is a dependency map, not a substitute for C5, C6, C10, or C11.

5.2 C7.2 The two-factor decomposition

The structural identity is

$$4\pi^2 = (2\pi)(2\pi).$$

Within the current framework, the two factors are interpreted as follows.

5.2.1 C7.2.1 Phase-convention factor

The first 2π enters because SFT counts one irreversible bit per **complete phase cycle**, so the bulk maintenance rate uses h, while Planck-scale geometry is conventionally written using $\hbar$:

$$h = 2\pi\hbar.$$

This is the linear-cycle versus angular-phase distinction. It is not a notational accident; it is a physical convention built into the Compton Maintenance Postulate and tested indirectly by the weak-field results.

5.2.2 C7.2.2 Boundary-versus-bulk normalization factor

The second 2π enters when one compares how boundary information capacity is counted and how bulk gravitational coupling is normalized. On the boundary side, one uses the Bekenstein–Hawking-style area count

$$I = \frac{A}{4\ell_P^2}.$$

On the bulk side, standard GR's cosmological and local field equations inherit the Einstein-side $8\pi G$ normalization.

The present framework identifies this as a structurally real mismatch between two normalization conventions. In the current master, this second factor is algebraically located and physically motivated, but still listed among the open foundational derivations when it comes to a deeper first-principles proof.

So the companion should state this plainly:

$$4\pi^2 = \underbrace{2\pi}_{h \text{ vs } \hbar} \cdot \underbrace{2\pi}_{\text{boundary vs bulk normalization}}.$$

5.3 C7.3 Appearance 1: Modified Friedmann ratio

The first appearance is the exact ratio between the SFT modified Friedmann equation and the standard flat-dust Friedmann equation:

$$\frac{H_{\text{SFT}}^2}{H_{\text{std}}^2} = \frac{1}{4\pi^2}.$$

This is derived explicitly in C5. It is the foundational appearance, because it is where the two 2π factors first become inseparable in the cosmological background law. Every later appearance is either a direct descendant of this result or a structurally parallel manifestation of the same counting mismatch.

Companion dependency: C5
Status: Derived

5.4 C7.4 Appearance 2: Galactic acceleration scale

The second appearance is the weak-field acceleration threshold. Since the Friedmann ratio lives at the level of H^2, the galactic acceleration threshold inherits its **square root**, so only one factor of 2π is exposed directly:

$$a_0 \sim \frac{cH_0}{2\pi}.$$

The SPARC sector then tests whether this is the right weak-field scale and whether the complete-cycle convention is favored over nearby alternatives. In the current framework, this is the appearance that provides the strongest empirical anchor for the phase-convention choice.

Companion dependency: C6
Status: Derived and empirically confirmed

5.5 C7.5 Appearance 3: Holographic capacity gap

The third appearance is the apparent mismatch between the total information capacity of the cosmological boundary and the total bulk processing history of the universe. In the master paper this is framed as an unsolved problem in standard thinking but as a bookkeeping artifact in SFT: the boundary and bulk counts are being written in inconsistent phase conventions. Once the counting conventions are made consistent, the apparent over-provisioning disappears structurally rather than requiring new physics.

This is not a separate theorem from Appearance 1. It is the same mismatch seen from the standpoint of integrated capacity rather than the background expansion law.

Companion dependency: C5 plus C3–4
Status: Derived in structural form

5.6 C7.6 Appearance 4: Late-universe structural gap

The fourth appearance is the late-universe structural gap between the matter-only SFT backbone and the observed expansion history. The master paper now frames this carefully: the matter-only background law undershoots the observed late-time expansion, and the gap is not to be filled by a free cosmological constant but by maintenance-regime dynamics. The size and structure of the gap are therefore constrained by the same backbone that generated the Friedmann ratio in the first place.

This appearance is not a separate algebraic derivation of $4\pi^2$; it is the **late-time dynamical consequence** of the same structural backbone once the active and passive maintenance channels are introduced.

Companion dependency: C10
Status: Structurally anchored, dynamically developed

5.7 C7.7 Appearance 5: Candidate CMB transfer splice

The fifth appearance is the early-universe candidate splice in the CMB transfer coefficient:

$$B_{\text{therm,candidate}} = \frac{q_T\, f_b\, f_r}{4\pi^2}.$$

This is the only appearance that the master explicitly keeps at candidate-derivation status. The numerical evidence is strong enough to justify inclusion:

- the filtered projection sweep identifies

$$B_{\text{therm}} \sim 3 \times 10^{-4},$$

- transfer-budget extraction says that coefficient is not mostly a detector artifact,
- partition tests favor a compound coefficient,
- and the $1/(4\pi^2)$ splice yields direct ballpark hits across mild parameter ranges.

But the unbroken symbolic theorem is still missing. So this appearance must remain explicitly marked as a candidate splice.

Companion dependency: C11
Status: Candidate splice / Open Theorem 3 boundary

5.8 C7.8 Dependency graph

The cleanest way to read the five appearances is as a dependency graph:

$$\text{Compton-cycle convention} + \text{boundary/bulk normalization} \longrightarrow \text{Appearance 1}$$

$$\text{Appearance 1} \longrightarrow \text{Appearance 2}$$

$$\text{Appearance 1} \longrightarrow \text{Appearance 3}$$

$$\text{Appearance 1} \longrightarrow \text{Appearance 4}$$

$$\text{Appearance 1 plus early-universe bridge/projection program} \longrightarrow \text{Appearance 5.}$$

This is the right way to avoid overclaiming. The appearances are related, but not all in the same way:

- 1–4 are on firmer ground,
- 5 is attached by a candidate splice.

5.9 C7.9 Same factor, extension of same factor, or coincident factor?

V10 is careful not to settle this prematurely, and the companion should keep that caution.

There are at least three logically distinct possibilities:

1. The fifth appearance is **the same** $4\pi^2$ in full physical identity.
2. It is a **consistent extension** of the same angular-versus-linear bookkeeping mechanism into the CMB sector.
3. It is a **numerically equal but physically distinct** factor.

The current framework does not yet prove which of these is true. That question belongs exactly where the master places it: at the boundary of Open Theorem 3.

5.10 C7.10 What this chapter has established

This chapter has established that the $4\pi^2$ recurrence is not a loose rhetorical motif. It has a concrete dependency structure:

- the foundational ratio appears in C5,
- its weak-field manifestation is audited in C6,
- its bookkeeping/capacity interpretation is anchored by C3–4 and C5,
- its late-time dynamical role is carried by C10,
- and its early-universe candidate extension is carried by C11.

That is enough for the companion. The detailed algebra belongs in the domain chapters themselves.

6. C9. Generic Forced-Transition Scaffolding

Master section mirrored: Section 9

Claim being derived: The maintenance process in SFT is not assumed to evolve smoothly under all conditions. It has viability conditions, and when those conditions fail, the system is forced into a new regime. The engine crash and the shatter are two instances of one generic thermodynamic mechanism.

Inputs / assumptions:

1. Landauer cost
2. Thermal-time interpretation of irreversible change
3. Finite holographic accommodation capacity
4. Maintenance process as a thermodynamically regulated system
5. Standard phase-transition scaffolding as the mildest mathematical language for regime change

Result: A generic forced transition is described by:

- a maintenance budget,
- a critical threshold,
- and an order parameter that emerges when the current regime becomes unsustainable.

The late-universe engine crash and the early-universe shatter are the two current applications of this scaffold.

Status: Structural scaffold / not a fully unified closed theorem.

2. C8. Holographic Count Equation and the Constrained Action

Master section mirrored: Section 8

Claim being derived: The SFT cosmological background law can be written as a holographic count equation over one Hubble settlement interval, and the same background law can be enforced through a constrained minisuperspace action.

Inputs / assumptions:

1. Flat FLRW background with scale factor $a(t)$ and Hubble parameter
$$H = \frac{\dot{a}}{a}$$
2. Hubble radius
$$R_H = \frac{c}{H}$$
3. Hubble-horizon area
$$A_H = 4\pi R_H^2 = 4\pi \left(\frac{c}{H}\right)^2$$
4. Hubble-volume energy content
$$E_H = \rho c^2 V_H, \qquad V_H = \frac{4\pi}{3}\left(\frac{c}{H}\right)^3$$
5. Planck area
$$\ell_P^2 = \frac{\hbar G}{c^3}$$
6. Linear-cycle convention
$$h = 2\pi\hbar$$
7. Pressureless dust continuity equation
$$\dot{\rho} + 3H\rho = 0$$

Result: The holographic count equation
$$\frac{A_H}{4\ell_P^2} = \frac{E_H}{hH}$$
reproduces
$$H^2 = \frac{2G\rho}{3\pi}.$$

The same background law can be enforced by the constrained minisuperspace action
$$S_{\text{eff}} = \int dt\, \Lambda(t) \left[\frac{A_H(a,H)}{4\ell_P^2} - \frac{E_H(a,\rho)}{hH}\right] + S_{\text{matter}},$$
with variation with respect to Λ imposing the count equation and the FLRW-reduced dynamics reproducing the same background relation.

Status: Derived at the FLRW constrained-minisuperspace level.

2.1 C8.1 Why the instantaneous rate-law attempt was structurally incomplete

Before the count equation was written in its present form, an earlier route tried to impose the bookkeeping balance as an instantaneous rate law,

$$\frac{d}{dt}\left(\frac{A_H}{4\ell_P^2}\right) = \gamma\,\dot{N}_{\text{tick}},$$

with γ an undetermined numerical factor and $\dot{N}_{\text{tick}}$ the instantaneous bulk ticking rate.

That formulation failed for a specific structural reason: it attempted to compare a boundary-capacity change and a bulk-processing rate at the same instant without specifying the causal interval over which the bulk processing is physically settled against the boundary that accommodates it. When this is done naively, the algebra leaves behind a free normalization factor that must be hand-set. In earlier exploratory work that factor was $\gamma = 3$, which was a signal that the formulation was missing a piece of geometry rather than discovering a new constant.

The count equation resolves this by replacing the instantaneous comparison with a one-Hubble-time settlement balance. Once that causal settlement interval is included, the missing factor emerges from the volume-to-area structure of the Hubble sphere rather than from a fitted coefficient.

2.2 C8.2 The one-Hubble-time settlement interval

The guiding idea is simple: the cosmological ledger cannot be settled instantaneously. Bulk processing in the Hubble volume must be compared to the boundary capacity that can causally accommodate it over the finite interval during which information can traverse the Hubble radius.

That interval is naturally

$$t_H = \frac{1}{H}.$$

This is the Hubble time, the characteristic causal timescale associated with the Hubble radius $R_H = c/H$. Over that interval, a signal moving at speed c can traverse a distance of order R_H. In flat FLRW, this makes the Hubble sphere the natural apparent-horizon-scale bookkeeping boundary for the minisuperspace treatment.

So the physically meaningful question is not: what is the instantaneous equality between bulk ticking and boundary growth? Instead, it is: over one Hubble settlement interval, how many bulk cycles occur, and how many boundary cells are available to accommodate them?

2.3 C8.3 Derivation of the holographic count equation

The count equation states that the number of available boundary cells equals the number of bulk Compton cycles accumulated over one Hubble time:

$$\boxed{\frac{A_H}{4\ell_P^2} = \frac{E_H}{hH}.}$$

The left-hand side is the Bekenstein-Hawking count of Planck-area cells on the Hubble horizon.

The right-hand side counts total bulk cycles:

- E_H/h is the total linear-cycle rate associated with the Hubble-volume energy content,
- multiplying by $1/H$ converts that rate into the number of cycles accumulated over one Hubble interval.

So the equation is a direct horizon-bookkeeping statement:

$$N_{\text{boundary}} = N_{\text{bulk ticks over one Hubble time}}.$$

2.4 C8.4 Explicit reduction of the count equation

Start from the left-hand side:

$$\frac{A_H}{4\ell_P^2} = \frac{4\pi(c/H)^2}{4\ell_P^2} = \frac{\pi c^2}{\ell_P^2 H^2}.$$

Now write the Hubble-volume energy:

$$E_H = \rho c^2 V_H = \rho c^2 \frac{4\pi}{3}\left(\frac{c}{H}\right)^3 = \frac{4\pi}{3}\rho\frac{c^5}{H^3}.$$

Therefore the right-hand side becomes

$$\frac{E_H}{hH} = \frac{1}{hH}\frac{4\pi}{3}\rho\frac{c^5}{H^3} = \frac{4\pi}{3}\rho\frac{c^5}{hH^4}.$$

So the count equation is

$$\frac{\pi c^2}{\ell_P^2 H^2} = \frac{4\pi}{3}\rho\frac{c^5}{hH^4}.$$

Now substitute

$$\ell_P^2 = \frac{\hbar G}{c^3} = \frac{hG}{2\pi c^3}.$$

Then

$$\frac{1}{\ell_P^2} = \frac{2\pi c^3}{hG}.$$

So the left-hand side becomes

$$\frac{\pi c^2}{\ell_P^2 H^2} = \frac{\pi c^2}{H^2}\cdot\frac{2\pi c^3}{hG} = \frac{2\pi^2 c^5}{hGH^2}.$$

Thus the count equation reads

$$\frac{2\pi^2 c^5}{hGH^2} = \frac{4\pi}{3}\rho\frac{c^5}{hH^4}.$$

Now cancel c^5 and h:

$$\frac{2\pi^2}{GH^2} = \frac{4\pi}{3}\frac{\rho}{H^4}.$$

Multiply both sides by GH^4:

$$2\pi^2 H^2 = \frac{4\pi}{3} G\rho.$$

Divide by $2\pi^2$:

$$H^2 = \frac{4\pi}{3}\frac{G\rho}{2\pi^2} = \frac{2G\rho}{3\pi}.$$

So the count equation reproduces exactly

$$\boxed{H^2 = \frac{2G\rho}{3\pi}.}$$

2.5 C8.5 Where the missing 3 comes from

In the count equation, the factor of 3 is not an extra constant. It comes from the Hubble-volume geometry through

$$V_H = \frac{4\pi}{3}\left(\frac{c}{H}\right)^3,$$

while the boundary capacity comes from the horizon area

$$A_H = 4\pi\left(\frac{c}{H}\right)^2.$$

The ratio between a sphere's volume and its surface area is what inserts the factor that the rate-law attempt had to encode by hand. The count equation does not discover $\gamma = 3$; it shows that γ was never fundamental in the first place.

2.6 C8.6 Why the count equation is the right variational object

The count equation is better than the algebraic master-equation substitution for one specific reason: it identifies the actual constraint being imposed on the minisuperspace system.

The constrained object is not a local field density. It is a horizon-scale balance:

$$\text{boundary cells} - \text{bulk ticks over one Hubble interval} = 0.$$

That is already the right shape for a constrained action principle. Instead of trying to write a local Lagrangian density for information maintenance, the FLRW problem can be written as a horizon-level constrained minisuperspace system, which is more natural for a holographic framework anyway.

2.7 C8.7 Construction of the constrained minisuperspace action

Introduce a Lagrange multiplier $\Lambda(t)$ and enforce the count equation as a constraint:

$$\boxed{S_{\text{eff}} = \int dt\, \Lambda(t)\left[\frac{A_H(a,H)}{4\ell_P^2} - \frac{E_H(a,\rho)}{hH}\right] + S_{\text{matter}}.}$$

Here:

- $a(t)$ is the FLRW scale factor,
- $H = \dot{a}/a$,
- $A_H(a, H)$ is the Hubble-horizon area,
- $E_H(a, \rho)$ is the Hubble-volume energy content,
- S_{matter} is the matter-sector minisuperspace action, whose role is to enforce the usual dust continuity law.

This is not a local covariant action for the full theory. It is a constrained minisuperspace action appropriate to the FLRW reduction. That limitation is important and is stated explicitly.

2.8 C8.8 Variation with respect to the Lagrange multiplier

The cleanest variation is the one with respect to $\Lambda(t)$:

$$\frac{\delta S_{\text{eff}}}{\delta \Lambda} = 0 \quad \Longrightarrow \quad \frac{A_H}{4\ell_P^2} - \frac{E_H}{hH} = 0.$$

So the Lagrange multiplier enforces exactly the holographic count equation:

$$\boxed{\frac{A_H}{4\ell_P^2} = \frac{E_H}{hH}.}$$

Since Section C8.4 already showed that this count equation yields

$$H^2 = \frac{2G\rho}{3\pi},$$

the action reproduces the same modified Friedmann background law.

This is the minimal derivation needed to justify the statement made in the master paper: the modified Friedmann law can be embedded in a standard constrained variational scaffold rather than appearing only as an algebraic coincidence.

2.9 C8.9 FLRW-reduced dynamics and the role of S_{matter}

To close the minisuperspace system, one also needs the matter continuity law. For pressureless dust,

$$\dot{\rho} + 3H\rho = 0.$$

In the constrained action picture, this is the role of the matter-sector term S_{matter}. At the FLRW background level, it is enough to state that the matter action is chosen so that the standard dust continuity equation holds.

So the constrained system is:

1. variation with respect to Λ gives the count equation,
2. the count equation gives the modified Friedmann law,
3. S_{matter} supplies the continuity equation.

That is sufficient to define the minisuperspace background dynamics.

2.10 C8.10 What is and is not being claimed here

This chapter proves the following:

- the holographic count equation
$$\frac{A_H}{4\ell_P^2} = \frac{E_H}{hH}$$
reproduces the modified Friedmann law exactly;
- the same count equation can be imposed as a constraint through a minisuperspace action;
- the resulting FLRW-reduced theory has a proper constrained variational scaffold.

This chapter does not claim:

- a full local covariant field-theory action for SFT,
- a derivation beyond the FLRW minisuperspace sector,
- or a completed nonlinear extension of the holographic constraint to arbitrary spacetimes.

Those are later research targets.

2.11 C8.11 Comparison of the four formulations

It is useful to summarize the four cosmological formulations now in play.

(1) Master equation
$$H = \frac{\dot{S}}{I}$$
This is the native thermodynamic identity of the framework.

(2) Algebraic Friedmann derivation Substituting the Compton maintenance rate and horizon capacity directly into the master equation gives
$$H^2 = \frac{2G\rho}{3\pi}.$$

(3) Holographic count equation
$$\frac{A_H}{4\ell_P^2} = \frac{E_H}{hH}$$

This is the causal one-Hubble-time settlement form of the same content.

(4) Constrained minisuperspace action

$$S_{\text{eff}} = \int dt\, \Lambda(t) \left[\frac{A_H}{4\ell_P^2} - \frac{E_H}{hH} \right] + S_{\text{matter}}.$$

These are not four unrelated proposals. They are four formulations of the same background content at increasing levels of mathematical structure.

2.12 C8.12 What this chapter has established

The correct way to promote the SFT background law into variational form is not through an instantaneous rate equation with a hand-set constant, but through a one-Hubble-time holographic count balance.

The count equation

$$\frac{A_H}{4\ell_P^2} = \frac{E_H}{hH}$$

is the natural causal bookkeeping statement of the framework, and it yields

$$\boxed{H^2 = \frac{2G\rho}{3\pi}}$$

with no free normalization.

That same count equation can be enforced through a constrained minisuperspace action,

$$\boxed{S_{\text{eff}} = \int dt\, \Lambda(t) \left[\frac{A_H}{4\ell_P^2} - \frac{E_H}{hH} \right] + S_{\text{matter}},}$$

which is the native FLRW variational scaffold of the theory.

6.1 C9.1 Why a generic scaffold is needed

The master paper's Sections 10 and 11 are two very different physical stories:

- late-time depletion of active astrophysical engines,
- early-time cooling through a thermal threshold.

If these were just placed side by side without a common formal language, the theory would look like it was applying two unrelated tricks. Section 9 prevents that. Its job is to state the **generic mechanism** once, so that the two epoch-specific sectors can be recognized as instances of one thermodynamic pattern rather than independent model additions.

The companion should preserve that role. It should not try to close the full unified theorem here; it should simply make the generic mechanism explicit.

6.2 C9.2 Maintenance budget

The generic object is a maintenance budget

$$\Lambda(X) = \frac{\Gamma_{\mathrm{req}}(X)}{\Gamma_{\mathrm{cap}}(X)},$$

where:

- X is a control variable or set of control variables,
- Γ_{req} is the maintenance demand of the current regime,
- Γ_{cap} is the maximum maintenance throughput the current regime can support.

The control variable X is intentionally generic:

- in the late universe, X is tied to structural exhaustion and channel depletion,
- in the early universe, X is tied to temperature.

The physical interpretation is simple:

- $\Lambda < 1$: the regime is viable,
- $\Lambda = 1$: the regime is critical,
- $\Lambda > 1$: the regime is insolvent and must reorganize.

That is the generic forced-transition condition.

6.3 C9.3 Critical threshold

The generic critical condition is therefore

$$\boxed{\Lambda(X_c) = 1.}$$

This is the moment when the current maintenance architecture becomes thermodynamically invalid. Importantly, the theory is not saying that every change in X produces a catastrophe. Most changes are absorbed adiabatically. The forced transition occurs only when the current regime literally cannot continue satisfying the local or global balance conditions.

That is what distinguishes a forced transition from ordinary slow evolution.

6.4 C9.4 Order-parameter scaffold

Near the critical threshold, the mildest standard thermodynamic language is Landau scaffolding.

Define a control parameter

$$\epsilon(X)$$

such that

$$\epsilon = 0 \quad \text{at} \quad X = X_c.$$

Then introduce an order parameter Ψ with the generic near-critical behavior

$$\boxed{\Psi \propto \sqrt{\epsilon} \qquad (\epsilon > 0).}$$

The point of this is not to commit the whole theory to one universal quartic free-energy form. The point is to give the transition a mathematically disciplined language:

- there is a pre-transition regime,
- there is a critical point,
- there is a post-transition regime with a new order parameter.

This is exactly how V10 uses Landau form in both sectors: as the mildest disciplined scaffolding, not as an overclaimed microscopic derivation.

6.5 C9.5 The late-universe engine crash as an instance

In the late universe, the control variable is effectively the balance between:

- the declining active channel,
- the rising passive structural channel.

The current maintenance regime becomes insolvent when the active engines can no longer service the accumulated maintenance burden at the required rate. The system is then forced into a structure-dominated regime, and the observable scar is the evolving effective dark-energy equation of state. That is the engine crash.

In the language of the generic scaffold:

- Γ_{req} is the maintenance demand implied by the cosmic ledger,
- Γ_{cap} is the active channel's servicing capacity under the current regime,
- the passive channel becomes dominant only after the crash.

The order parameter is not a textbook magnetization here; it is the degree to which the maintenance burden has shifted from active processing to passive structural occupancy.

6.6 C9.6 The early-universe shatter as an instance

In the early universe, the control variable is temperature. The shared radiation-dominated maintenance regime becomes insolvent as the universe cools and independent massive-particle clocks begin to demand separate bookkeeping. The current regime can no longer maintain collective solvency, so the system is forced into an independent-clock regime. The observable scar is the relic phase perturbation field and its later CMB imprint.

In the language of the generic scaffold:

- Γ_{req} is the maintenance burden associated with the emerging independent-clock population,
- Γ_{cap} is the shared thermal regime's capacity to handle that burden collectively.

The order parameter is then the degree of independent-clock structure in the post-shatter regime.

6.7 C9.7 Structural unification versus formal unification

This distinction is crucial and V10 makes it explicitly.

What the framework has now is **structural unification**:

- both sectors use the same generic transition logic,
- both sectors use threshold crossing,
- both sectors admit Landau-form scaffolding,
- both sectors leave a geometric scar.

What it does **not** yet have is a single closed unified equation in which the late-time engine crash and the early-time shatter appear as two different solutions of one master transition law. V10 names that as an open theorem, and the companion should preserve that exact boundary.

So the correct statement is:

$$\boxed{\text{One mechanism in structural form, not yet one theorem in closed form.}}$$

6.8 C9.8 What this chapter has established

This chapter has established the generic transition scaffold used by both C10 and C11:

1. a maintenance budget
$$\Lambda = \Gamma_{\text{req}}/\Gamma_{\text{cap}},$$
2. a critical condition
$$\Lambda = 1,$$
3. a post-critical order parameter
$$\Psi \propto \sqrt{\epsilon}.$$

That is the minimal formal machinery needed to understand the engine crash and the shatter as members of one family of thermodynamic events.

A. Appendix A. Constants, Values, and Unit Checks

Purpose: Collect the constants, numerical values, and dimensional identities used repeatedly in the companion, so that the main chapters do not have to keep re-deriving them.

A.1 A.1 Fundamental constants

The companion uses the following constants throughout:

$$c = \text{speed of light}, \qquad G = \text{Newtonian gravitational constant}, \qquad h = \text{linear Planck constant}, \qquad \hbar = \frac{h}{2\pi}, \qquad k$$

Where useful, the reduced and linear Planck constants are related explicitly by

$$h = 2\pi\hbar.$$

This relation is load-bearing because the framework counts one irreversible bit per **complete phase cycle**, not per radian of phase advance. That convention is one of the two structural sources of the recurring 2π factors in the theory.

A.2 A.2 Planck scales

The Planck length and Planck area are

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \qquad \ell_P^2 = \frac{\hbar G}{c^3} = \frac{hG}{2\pi c^3}.$$

Numerically,

$$\ell_P \approx 1.616 \times 10^{-35}\ \text{m}, \qquad \ell_P^2 \approx 2.61 \times 10^{-70}\ \text{m}^2.$$

3. C10. Cosmological Sector: Full Ladder Derivation and TNG300 Methodology

C10.1–C10.4 Four-State Ladder and Static Projection Corrections

Master section mirrored: Section 10

Claim being derived: The late-universe cosmological sector can be represented by a four-state cosmic-web ladder whose ideal weights are

$$w_{\text{void}} = 4\pi^2, \qquad w_{\text{wall}} = (2\pi)^{4/3}, \qquad w_{\text{filament}} = (2\pi)^{2/3}, \qquad w_{\text{halo}} = 1,$$

and whose extended-structure corrections are

$$s_{\text{wall}} = \frac{1}{3}, \qquad s_{\text{filament}} = \frac{2}{3}.$$

Inputs / assumptions:

1. Four cosmic-web environments: void, wall, filament, halo.
2. Maintenance burden is transmitted to the cosmological boundary only along directions not already locally absorbed by organized gravitational curvature.
3. Axis factorization is valid at leading order.
4. The fully unscreened limit must reproduce the same structural $4\pi^2$ factor that appears elsewhere in the framework.
5. Extended walls and filaments require projection softening relative to the ideal factorized ladder.

Result: The ideal ladder follows from the three structural axioms, and the static softening factors follow from second moments of spherical-screen projection:

$$\langle \cos^2\theta \rangle_{\mathbb{S}^2} = \frac{1}{3}, \qquad \langle \sin^2\theta \rangle_{\mathbb{S}^2} = \frac{2}{3}.$$

Status: Structural derivation for the ideal ladder; projection-softening derivation for the static corrections.

3.1 C10.1 Cosmic-web state definitions

The cosmological sector of SFT decomposes the late-time universe into four gravitational environments:

- voids: essentially unscreened regions with negligible local gravitational infrastructure,
- walls: two-dimensional sheet-like structures,

- filaments: one-dimensional elongated structures,
- halos: fully virialized three-dimensional collapsed structures.

The physical meaning of the decomposition is not merely morphological. Each environment represents a different way in which local curvature can absorb the Compton maintenance burden before that burden is transmitted to the cosmological boundary. Halos absorb almost everything locally, voids transmit almost everything to the boundary, and walls and filaments occupy intermediate screening states because they are organized only along some subset of spatial directions.

To formalize this, define a dimensionless transmitted-maintenance weight

$$w_{\mathrm{env}},$$

interpreted as the boundary-directed maintenance burden of a unit mass element in environment "env," normalized to the fully screened halo case.

At the qualitative level:

- w_{halo} is smallest,
- w_{void} is largest,
- w_{filament} and w_{wall} lie between them.

The goal of this chapter is to derive the exact ideal ladder and then derive the leading static corrections for the extended intermediate states.

3.2 C10.2 Three structural axioms of the ladder

The ladder is built from the three structural axioms stated in V10.

3.2.1 C10.2.1 Axiom C1: volume-weighted neutrality

The total maintenance demand of the universe, averaged over all environments, must equal the maintenance demand computed at the mean density. In other words, the cosmic web redistributes where maintenance is locally absorbed and where it is boundary-directed, but it does not create or destroy total maintenance demand.

If f_{env} denotes the fraction of mass or volume assigned to environment "env," then the ladder weights must satisfy a global consistency condition of the schematic form

$$\sum_{\mathrm{env}} f_{\mathrm{env}}\, w_{\mathrm{env}}$$

being the correct cosmological effective burden measure relative to the chosen bookkeeping convention.

This axiom does not by itself fix the ladder values, but it tells us they must be interpretable as redistribution weights rather than fitted phenomenological labels.

3.2.2 C10.2.2 Axiom C2: axis factorization

The key structural input is that transmission to the cosmological boundary factors over the independent spatial axes that remain open.

This is the geometric content of the environment classification:

- halo: 0 open axes
- filament: 1 open axis
- wall: 2 open axes
- void: 3 open axes

Equivalently, if a single open axis contributes a fixed geometric factor g, then the ideal transmitted burden scales as

$$w(N_{\text{open}}) = g^{N_{\text{open}}}.$$

This is the leading-order factorization assumption: each open axis contributes independently and multiplicatively to the boundary-directed maintenance burden.

3.2.3 C10.2.3 Axiom C3: consistency with the structural identity

The fully unscreened limit must reproduce the same structural factor that appears throughout the framework:

$$4\pi^2 = (2\pi)^2.$$

Since a void has all three axes open, the ideal void weight must be

$$w_{\text{void}} = 4\pi^2.$$

At the opposite extreme, a fully screened halo defines the normalization:

$$w_{\text{halo}} = 1.$$

So if the single-axis geometric factor is g, then

$$g^3 = 4\pi^2.$$

Therefore

$$g = (4\pi^2)^{1/3} = (2\pi)^{2/3}.$$

That is the per-open-axis transmission factor.

3.3 C10.3 Derivation of the ideal four-state ladder

With the three axioms in place, the ideal ladder follows immediately.

Let

$$g = (2\pi)^{2/3}.$$

Then the transmitted burden for a region with N_{open} open axes is

$$w(N_{\text{open}}) = g^{N_{\text{open}}}.$$

Now apply this to the four environments.

3.3.1 C10.3.1 Halo

A fully virialized halo is taken as completely screened at the level of the ideal ladder, so

$$N_{\text{open}} = 0.$$

Therefore

$$w_{\text{halo}} = g^0 = 1.$$

3.3.2 C10.3.2 Filament

A filament is open along one effective axis, so

$$N_{\text{open}} = 1.$$

Therefore

$$w_{\text{filament}} = g = (2\pi)^{2/3}.$$

3.3.3 C10.3.3 Wall

A wall is open along two effective axes, so

$$N_{\text{open}} = 2.$$

Therefore

$$w_{\text{wall}} = g^2 = (2\pi)^{4/3}.$$

3.3.4 C10.3.4 Void

A void is open along all three axes, so

$$N_{\text{open}} = 3.$$

Therefore

$$w_{\text{void}} = g^3 = (2\pi)^2 = 4\pi^2.$$

So the ideal ladder is

$$\boxed{w_{\text{void}} = 4\pi^2, \qquad w_{\text{wall}} = (2\pi)^{4/3}, \qquad w_{\text{filament}} = (2\pi)^{2/3}, \qquad w_{\text{halo}} = 1.}$$

This is the ladder stated in V10. The master paper presents it as the natural ladder consistent with the three structural axioms, while leaving the stronger uniqueness claim to the companion-level discussion.

3.3.5 C10.3.5 Interpretation of the ladder

The ladder is best understood as an ideal factorized burden hierarchy:

- the void sits at full unscreened transmission,
- the halo sets the fully screened normalization,
- filament and wall are intermediate dimensional states.

This is the leading-order structural closure.

But these are ideal weights. Real walls and filaments are extended objects with finite thickness and anisotropic exposure to the spherical boundary. So the factorized ladder captures the dimensional count, but not yet the projection asymmetry of extended structures. That is what the static correction factors are for.

3.4 C10.4 Static projection corrections from spherical-screen integrals

The ideal ladder treats each open axis as contributing the same multiplicative factor. That is the correct leading-order count, but it does not yet account for the fact that walls and filaments are extended geometries rather than pointlike or isotropic sources.

The needed correction is a projection softening:

- a wall only receives the normal component of the spherical-screen burden,
- a filament receives the transverse two-dimensional component.

This is the same projection-integral mechanism that the weak-field sector uses for the galactic λ values.

3.4.1 C10.4.1 Wall correction

Model a wall locally as a flat sheet with unit normal $\hat{n}$. Let $\hat{r}$ be a direction on the spherical screen, with polar angle θ measured relative to $\hat{n}$. The component of exposure normal to the sheet is proportional to

$$\cos\theta.$$

For a second-moment burden measure, the relevant average is

$$\langle \cos^2 \theta \rangle_{\mathbb{S}^2}.$$

Using the normalized spherical average,

$$\langle f(\theta, \phi) \rangle_{\mathbb{S}^2} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} f(\theta, \phi) \sin\theta \, d\theta \, d\phi,$$

we compute

$$\langle \cos^2 \theta \rangle_{\mathbb{S}^2} = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi} \cos^2 \theta \, \sin\theta \, d\theta \, d\phi.$$

The ϕ-integral gives 2π, so

$$\langle \cos^2 \theta \rangle_{\mathbb{S}^2} = \frac{1}{2} \int_0^{\pi} \cos^2 \theta \, \sin\theta \, d\theta.$$

Let $u = \cos\theta$, so $du = -\sin\theta \, d\theta$. Then

$$\int_0^{\pi} \cos^2 \theta \, \sin\theta \, d\theta = \int_1^{-1} u^2 (-du) = \int_{-1}^{1} u^2 \, du = \frac{2}{3}.$$

Therefore

$$\langle \cos^2 \theta \rangle_{\mathbb{S}^2} = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

So the wall softening factor is

$$\boxed{s_{\text{wall}} = \frac{1}{3}.}$$

The corrected wall weight is then

$$w_{\text{wall}}^{\text{corr}} = s_{\text{wall}} \, w_{\text{wall}} = \frac{1}{3} (2\pi)^{4/3}.$$

3.4.2 C10.4.2 Filament correction

Model a filament as an extended one-dimensional object oriented along axis $\hat{e}$. What matters is not the longitudinal projection but the transverse exposure, since the filament is already organized along its long axis.

The squared transverse fraction is

$$1 - \cos^2 \theta = \sin^2 \theta,$$

where θ is measured relative to the filament axis.

So the relevant second moment is

$$\langle \sin^2 \theta \rangle_{\mathbb{S}^2}.$$

But since

$$\sin^2 \theta = 1 - \cos^2 \theta,$$

we immediately get

$$\langle \sin^2\theta \rangle_{\mathbb{S}^2} = 1 - \langle \cos^2\theta \rangle_{\mathbb{S}^2} = 1 - \frac{1}{3} = \frac{2}{3}.$$

So the filament softening factor is

$$\boxed{s_{\text{filament}} = \frac{2}{3}.}$$

The corrected filament weight is therefore

$$w_{\text{filament}}^{\text{corr}} = s_{\text{filament}}\, w_{\text{filament}} = \frac{2}{3}(2\pi)^{2/3}.$$

3.4.3 C10.4.3 Why these are the right static corrections

These factors are not arbitrary softening parameters. They are exactly the rotational second moments of a spherical burden measure projected onto one normal direction for a sheet and the complementary transverse plane for a filament.

That is why they are so geometrically clean:

$$\frac{1}{3}, \qquad \frac{2}{3}.$$

They are the standard isotropic second moments of one Cartesian component and its two-dimensional complement on the sphere.

This also explains why they are the natural first correction to the factorized ladder:

- the ladder counts how many axes are open,
- the static corrections measure how strongly a spherical boundary field projects onto those open directions.

3.4.4 C10.4.4 Structural unification with the galactic sector

The same projection logic appears in the weak-field sector, where different lower-dimensional geometries projected against a spherical-screen structure produce the benchmark λ values used in the SPARC analysis.

That means the cosmological corrections

$$s_{\text{wall}} = \frac{1}{3}, \qquad s_{\text{filament}} = \frac{2}{3}$$

should be read as the large-scale analogues of the galactic projection benchmarks:

- same mathematical object,
- different physical geometry,
- different observational sector.

3.4.5 C10.4.5 Corrected ladder summary

Putting everything together, the static-corrected cosmological weights are

$$w_{\text{void}}^{\text{corr}} = 4\pi^2, \qquad w_{\text{wall}}^{\text{corr}} = \frac{1}{3}(2\pi)^{4/3}, \qquad w_{\text{filament}}^{\text{corr}} = \frac{2}{3}(2\pi)^{2/3}, \qquad w_{\text{halo}}^{\text{corr}} = 1.$$

The void and halo endpoints are unchanged because they are already the ideal unscreened and ideal screened extremes. The extended intermediate states are softened by the spherical projection geometry.

3.4.6 C10.4.6 What has been established so far

This first part of the cosmological companion chapter has established three things:

1. The ideal four-state ladder follows from the three structural axioms:

$$w_{\text{void}} = 4\pi^2, \quad w_{\text{wall}} = (2\pi)^{4/3}, \quad w_{\text{filament}} = (2\pi)^{2/3}, \quad w_{\text{halo}} = 1.$$

2. The static projection corrections follow from spherical second moments:

$$s_{\text{wall}} = \frac{1}{3}, \qquad s_{\text{filament}} = \frac{2}{3}.$$

3. The same projection-integral logic structurally unifies the cosmological and galactic sectors.

C10.5–C10.16 Channel Construction, Historical Pipeline, Corrected Chain, and Audit Status

Master section mirrored: Section 10

Claim being derived: The late-universe cosmological sector is computed from an evolving two-channel history extracted from TNG300 and interpreted through the four-state ladder. The historical pipeline produced the quoted DESI-facing CPL pair

$$(w_0, w_a) = (-0.837, -0.615),$$

with phantom crossing near

$$z_{\text{cross}} \approx 0.465,$$

while the corrected framework-level derivation reconstructs the equation of state from the expansion history $H(z)$ and is currently under explicit audit for full quantitative reconciliation.

Inputs / assumptions:

1. Four-state ladder and static corrections from C10.1–C10.4
2. TNG300 halo catalogs, star-formation histories, and black-hole accretion histories
3. Active channel = ongoing irreversible astrophysical transactions

4. Passive channel = accumulated virialized gravitational architecture
5. Historical numerical pipeline and corrected $H(z) \to w_{\rm DE}(z)$ reconstruction are both retained, with distinct epistemic status.

Result: The cosmological sector is controlled by the time evolution of the active and passive channels. The historical chain yields the quoted CPL pair and crossing redshift. The corrected formal chain is

$$H(z) = \frac{\dot{S}_{\rm total}(z)}{I(z)}, \qquad \Omega_{\rm DE}(z) = E(z)^2 - \Omega_{m0}(1+z)^3, \qquad w_{\rm DE}(z) = -1 + \frac{1+z}{3\,\Omega_{\rm DE}(z)}\frac{d\Omega_{\rm DE}}{dz},$$

and its full rerun against the original channel history is a V10.1 audit item.

Status: Historical empirical pipeline established; corrected derivation established at the formula level; full numerical reconciliation under audit.

3.5 C10.5 Structural unification with the galactic projection-integral machinery

The static corrections

$$s_{\rm wall} = \frac{1}{3}, \qquad s_{\rm filament} = \frac{2}{3}$$

should not be read as isolated cosmological parameters. They arise from the same spherical-screen second-moment logic used in the weak-field sector, where the galactic morphology dependence is captured by the two benchmark λ values. One projection-integral mechanism is being applied at two different scales, with the geometry of the target structure determining which second moments survive.

At galactic scales, the lower-dimensional structures are disk-like or bulge-like baryonic configurations. At cosmological scales, the lower-dimensional structures are walls and filaments in the cosmic web. The companion therefore treats these as members of one family of projection problems:

- lower-dimensional geometry embedded in a spherical-screen burden field,
- second-moment projection onto the directions not already screened by local organization,
- resulting numerical factors determined by dimensionality and angular anisotropy.

This unification remains a structural theorem target, not yet a fully closed theorem. The goal is a single projection-integral theorem whose different limits reproduce

$$3\pi/8, \quad 4/3, \quad 1/3, \quad 2/3$$

from one underlying construction rather than from independent calculations.

3.6 C10.6 Active channel construction

The active processing channel is the rate of ongoing irreversible astrophysical transactions. In the V10 cosmology sector, this channel is built from two classes of processes:

- black-hole accretion,
- star formation.

These are the processes in which matter is actively converting gravitational potential energy, nuclear fuel, or accretion power into radiation, jets, or binding-energy changes. Within the SFT language, they are the dominant transactional contributions to the late-time information-production rate. They are not merely inventories of already-existing structure; they are ongoing irreversible events.

In the historical pipeline, the black-hole contribution was computed from accretion histories using a Bekenstein-Hawking-style scaling, while the stellar contribution was computed from the simulation's star-formation histories using a Kennicutt-Schmidt-style conversion. The important dynamical fact is the time evolution:

- the active channel peaks at moderate redshift,
- then declines sharply toward the present epoch,
- with an approximately 70% crash between the mid-redshift peak regime and the present epoch,
- and the steepest decline centered near

$$z \approx 0.465.$$

This crash is the empirical backbone of the engine-crash interpretation introduced in Section 9 and used as the dynamical source of the phantom crossing in Section 10.

3.7 C10.7 Passive channel construction

The passive structural capacity channel measures not ongoing transactions but accumulated gravitational organization. In the V10 and V8/V9 cosmology pipeline, this is represented by the virialized halo mass

$$M_{200c}$$

summed across all halos in each snapshot and then interpreted through the four-state ladder or its earlier bimodal predecessor.

The physical distinction between the active and passive channels is essential:

- the active channel is the cost of doing work now,
- the passive channel is the cost of maintaining already-built screening infrastructure.

This is why the passive channel is framed as the accumulated gravitational architecture that has already been built and is now maintaining itself. It is not the same thing as generic mass inventory. The channel is intended to track organized, virialized structure that can absorb local maintenance burdens, not all matter indiscriminately.

The passive channel grows monotonically toward low redshift as structure formation proceeds. So the late-time cosmological picture is not one of dark energy switching on from nowhere, but one of a declining active engine and a rising passive structural burden whose changing balance drives the effective equation-of-state evolution.

3.8 C10.8 TNG300 extraction methodology

The numerical history used in the historical pipeline comes from the IllustrisTNG TNG300 simulation. SFT uses it as a calibrated source of empirically valid gravitational structure and baryonic processing histories rather than as an endorsement of dark-matter ontology.

V10 states that the historical DESI-facing cosmology result was obtained from ten snapshots spanning

$$z = 2 \text{ to } z = 0.$$

The active and passive channels were extracted at those snapshots, then propagated through the cosmology pipeline to produce the quoted CPL pair and phantom-crossing location. This is the historical numerical backbone that the companion must preserve, even though the corrected derivation chain now changes how the reconstruction should ideally be expressed mathematically.

So the methodology is:

1. Select the TNG300 snapshots in the chosen redshift interval.
2. Extract active-channel observables:
   - black-hole accretion history,
   - star-formation history.
3. Extract passive-channel observable:
   - virialized halo mass M_{200c} history.
4. Weight or interpret the structural sector through the SFT screening ladder.
5. Construct the late-time effective cosmology pipeline.

3.9 C10.9 Structural-specificity test: M_{200c} versus M_{halo}

One of the most load-bearing empirical diagnostics in the cosmological sector is the structural-specificity test. The passive channel succeeds only when it is tied to virialized gravitational organization, not when it is replaced by generic halo inventory.

The comparison is:

- passive channel built from M_{200c}: succeeds,
- passive channel built from raw M_{halo}: fails.

In the historical pipeline, replacing M_{200c} with M_{halo} shifts the equation of state to approximately

$$w_0 = -1.065,$$

overshooting the phantom divide and missing the observed value. Within SFT, the quantity that matters is not the total amount of matter but the amount of matter that has been gravitationally organized into effective screening infrastructure.

This is a major discriminator because the passive channel is supposed to be a physical measure of organized maintenance burden, not a generic phenomenological reservoir.

3.10 C10.10 Dynamical reservoir ablation

The second load-bearing empirical comparison is the dynamical reservoir ablation. The question is: does the passive structural reservoir need to evolve dynamically, or is a static reservoir sufficient?

The answer from the historical pipeline is that a static reservoir fails badly. When the time-derivative contribution of the maintenance budget is removed — that is, when the passive channel is treated as static rather than evolving — the DESI likelihood collapses by between roughly 36 and 1263 log-units depending on the dataset combination.

This matters because it shows that the passive channel is not merely a late-time offset or nuisance correction. Its evolution is doing real dynamical work. The phantom-like behavior is not generated by the existence of structure alone, but by the historical change in the maintenance balance between the declining active channel and the rising passive one.

3.11 C10.11 Historical cosmology pipeline

The historical cosmology pipeline is the chain that produced the headline late-time result quoted throughout V9 and V10:

$$w_0 = -0.837, \qquad w_a = -0.615,$$

with phantom crossing at approximately

$$z_{\text{cross}} = 0.465.$$

V10 names this chain explicitly as historical. It does not erase the result, but it distinguishes the established numerical output from the mathematically corrected reconstruction now being developed alongside it.

What matters for the companion is to preserve the pipeline's role clearly:

- it used the active and passive channels extracted from TNG300,
- it produced the DESI-facing CPL pair,
- it converged stably across many scripts and intermediate checks,
- and it is the empirical numerical output that anchors the late-time cosmology claims in the current framework.

3.12 C10.12 Corrected framework-level derivation chain

The mathematically corrected chain starts from the expansion history, not directly from a one-line formula for w. This is the key fix.

3.12.1 Step 1: reconstruct the expansion history

The corrected reconstruction begins with

$$H(z) = \frac{\dot{S}_{\text{total}}(z)}{I(z)}.$$

This uses the same underlying channel history, but interprets it through the master equation directly rather than through the older working w-mapping.

3.12.2 Step 2: reconstruct the total effective equation of state

For a flat FLRW expansion history, the total effective equation-of-state parameter is

$$w_{\text{eff}}(z) = -1 + \frac{1+z}{3}\frac{d\ln H^2}{dz}.$$

This is the standard FLRW reconstruction identity, applied to the SFT expansion history. The load-bearing point is the dependence on the time evolution of the maintenance balance, not merely on the instantaneous ratio $\dot{S}/I$.

3.12.3 Step 3: isolate the dark-energy residual sector

To compare to DESI's CPL dark-energy constraints, one must subtract the matter contribution and define

$$\Omega_{\text{DE}}(z) = E(z)^2 - \Omega_{m0}(1+z)^3, \qquad E(z)^2 = \frac{H(z)^2}{H_0^2}.$$

Then the reconstructed dark-energy sector is

$$\boxed{w_{\text{DE}}(z) = -1 + \frac{1+z}{3\,\Omega_{\text{DE}}(z)}\frac{d\Omega_{\text{DE}}}{dz}.}$$

This is the formula implemented in the corrected audit script:

$$H(z) \to E(z)^2 \to \Omega_{\text{DE}}(z) \to w_{\text{DE}}(z).$$

This corrected chain is the mathematically preferred derivation because it is dimensionally clean and follows the standard logic of reconstructing an effective fluid from an expansion history.

3.13 C10.13 Historical-versus-corrected audit note

The quoted CPL pair belongs to the historical cosmology pipeline. The corrected $H(z) \to w_{\text{DE}}(z)$ chain is the mathematically preferred framework-level derivation. A first coarse rerun, using only a five-point channel-history summary, preserved the crossing scale but shifted the fitted CPL coefficients strongly:

$$w_0^{\text{fit}} \approx -2.154, \qquad w_a^{\text{fit}} \approx 4.863, \qquad z_{\text{cross}} \approx 0.477.$$

So the audit conclusion is not that the old result is invalid, but rather:

- the corrected derivation is the right formal chain,
- the old numerical result is the established empirical output of the historical pipeline,

- full reconciliation is still pending.

This is why V10 flags the issue as a V10.1 audit item. The master paper is explicit that the qualitative crossing behavior appears stable while the precise CPL coefficients remain under reconciliation.

3.14 C10.14 Phantom-crossing redshift extraction

One of the most encouraging pieces of overlap between the historical and corrected chains is the crossing redshift.

The historical cosmology pipeline gives phantom crossing at approximately

$$z_{\text{cross}} \approx 0.465.$$

The first coarse corrected rerun gave

$$z_{\text{cross}} \approx 0.477.$$

These are close enough to suggest that the dynamical transition timing is likely tied to a real feature of the channel history rather than being an artifact of one specific reconstruction formula. That does not prove full numerical equivalence, but it is a meaningful stability test for the structural story: the engine crash is occurring in the same redshift neighborhood in both chains.

For the companion, this means the crossing redshift should be treated as a qualitatively stable observable across current formulations, while the exact CPL coefficients remain an audit-sensitive output.

3.15 C10.15 CMB compatibility computation details

V10 keeps the late-time CMB compatibility result as part of the cosmological sector:

- propagated through CLASS as a CPL parameterization,
- Planck 2018 temperature power spectrum residual

  $$0.078\% \text{ RMS}$$

  over multipoles

  $$\ell = 2\text{–}2500.$$

Earlier framework text also gives the multipole-by-multipole breakdown: the acoustic peaks are preserved at sub-percent level, while the only noticeable residual is the low-ℓ ISW sector, where cosmic variance is already large.

The physical reason given by the framework is simple: at the CMB epoch there are no virialized halos, no late-time void network, and no mature screening infrastructure, so the late-time SFT modifications are naturally suppressed. In other words, the cosmological sector is constructed specifically to modify the late universe while leaving the early acoustic structure essentially standard.

The detailed CLASS settings, parameter priors, and exact implementation choices should go in the repro appendix, but the companion chapter should record the load-bearing fact:

$$\text{RMS residual} = 0.078\%.$$

3.16 C10.16 What this chapter has established

This chapter has established the following:

1. The ideal four-state ladder and the static projection corrections provide the structural cosmological weighting scheme:

$$w_{\text{void}} = 4\pi^2, \quad w_{\text{wall}}^{\text{corr}} = \frac{1}{3}(2\pi)^{4/3}, \quad w_{\text{filament}}^{\text{corr}} = \frac{2}{3}(2\pi)^{2/3}, \quad w_{\text{halo}} = 1.$$

2. The late-time cosmological history is built from two physically distinct channels: active astrophysical transactions and passive virialized structure.

3. The passive channel is structurally specific:

$$M_{200c} \text{ succeeds where } M_{\text{halo}} \text{ fails,}$$

   showing that organized gravitational architecture, not raw inventory, is the relevant quantity.

4. The passive channel must be dynamical, not static; otherwise the DESI likelihood collapses catastrophically.

5. The quoted CPL pair

$$(-0.837, -0.615)$$

   belongs to the established historical cosmology pipeline. The corrected framework-level derivation is

$$H(z) \to \Omega_{\text{DE}}(z) \to w_{\text{DE}}(z),$$

   and full numerical reconciliation of the two is explicitly under audit for V10.1.

6. The crossing redshift appears to be the more stable quantity across current chains than the exact CPL coefficients, which is encouraging for the dynamical interpretation of the engine crash.

So the cosmological sector should currently be read as:

- structurally derived at the ladder/channel level,
- historically empirically anchored at the CPL-output level,
- mathematically corrected at the framework-derivation level,
- and under targeted audit for complete chain reconciliation.

C11. Early-Universe Sector and Candidate CMB Transfer Splice

Companion Derivations to Solvency Field Theory (V10)

Master section mirrored: Section 11

Claim being derived: A relic phase perturbation generated by the early-universe shatter can project from a 3D source field onto the 2D last-scattering surface in a way that is geometrically viable, provided the observable is not the naked Laplacian of the phase field but a filtered thin-shell readout with a compound transfer coefficient. The required transfer coefficient is approximately

$$B_{\text{therm}} \sim 3 \times 10^{-4},$$

is not mostly a detector artifact, and is best represented at the current stage by the compound form

$$B_{\text{therm}} = q_T f_b f_r \varepsilon,$$

with a candidate splice

$$\varepsilon_{\text{candidate}} = \frac{1}{4\pi^2}.$$

Inputs / assumptions:

1. Section 9's forced-transition framework applies to the early universe as a thermal transition.
2. The cosmological lag variable is
$$\chi = \ln n,$$
where n is the lag refractive index of the information-maintenance ledger.
3. The screening fraction is
$$U_{\text{screen}} = 1 - F,$$
where F is the screened-volume fraction.
4. The relic phase field Φ is the early-universe phase variable whose perturbations project into the observed CMB anisotropy.
5. The mildest working bridge is the identity-like bridge stated in V10.

Result: The leading early-universe bridge is

$$\delta\chi = \delta\Phi, \qquad \bar{\Phi}(t) = \ln a(t), \qquad \dot{\bar{\Phi}} = H, \qquad U_{\text{screen}} \approx \alpha\chi.$$

This bridge is bounded by recombination-era amplitudes and supports a mild Landau-style shatter

regime with shallow temperature dependence and characteristic temperatures in the 5000–10000 K neighborhood.

Status: Candidate derivation / structurally viable / quantitatively constrained.

1. Role of the early-universe chapter

The early-universe sector is not at the same maturity level as the late-universe cosmological sector. Its job is not to claim a finished theorem, but to show that the forced-transition principle has a plausible and quantitatively constrained early-universe realization, and that this realization has a candidate link to the same $4\pi^2$ structural identity that appears elsewhere in the theory.

The right register is therefore:

- structurally viable,
- quantitatively bounded,
- not yet closed as an unbroken theorem.

2. The minimal identity-like bridge

The leading bridge between the late-time screening language and the relic phase field is

$$\delta\chi = \delta\Phi, \qquad \bar{\Phi}(t) = \ln a(t), \qquad \dot{\bar{\Phi}} = H, \qquad U_{\text{screen}} \approx \alpha\chi.$$

2.1. The perturbative identification

The perturbation in the lag variable is identified directly with the perturbation in the relic phase field:

$$\delta\chi = \delta\Phi.$$

This is the mildest nontrivial bridge. It says the cosmological lag variable and the relic phase perturbation are two epoch-dependent descriptions of the same underlying maintenance mismatch.

2.2. The background phase clock

The background relic phase is taken to be

$$\bar{\Phi}(t) = \ln a(t).$$

Differentiating gives

$$\dot{\bar{\Phi}} = \frac{d}{dt} \ln a = \frac{\dot{a}}{a} = H.$$

So the background phase rate is the FLRW count-equation clock itself.

2.3. The linear bridge to screening

The screening fraction is taken to be linear in the lag variable:

$$U_{\text{screen}} \approx \alpha\chi.$$

The coefficient α is the leading bridge coefficient. The linear form is adopted because it is the mildest possible nontrivial closure.

3. Why this is the right minimal bridge

The bridge is "identity-like" rather than "fully derived" for a reason. The right question is not whether the early-universe field has already been derived from first principles in full. It has not. The right question is: what is the mildest consistent bridge that lets the cosmological screening sector talk to the relic phase field and survive the first quantitative tests? At the current state of the theory, the answer is the identity-like bridge above.

4. Recombination-era mildness bounds

The identity-like bridge is not unconstrained. The observed CMB anisotropy scale bounds the size of the allowed relic perturbation, and under the bridge this directly bounds the size of $\delta\chi$ and therefore the size of the bridge coefficient α in

$$U_{\text{screen}} \approx \alpha\chi.$$

The outcome of the mildness scans is that the bridge coefficient lands in a mild range rather than requiring pathological values.

5. The shatter as a Landau-form phase transition

The early-universe shatter is modeled as the failure of a shared maintenance regime when a thermal control parameter crosses a critical threshold.

Write a temperature-dependent coherence overhead

$$\Lambda(T) = \eta \left(\frac{T_\star}{T}\right)^m,$$

with

- $T_\star$ a characteristic temperature scale,
- m a shallow exponent,
- η a dimensionless normalization.

Define the Landau-form control parameter

$$\epsilon(T) = \eta \left(\frac{T_\star}{T} - 1\right).$$

Then the equilibrium order parameter is taken to scale as

$$\Phi_{\mathrm{eq}}(T) \propto \sqrt{\epsilon(T)} \qquad (\epsilon > 0).$$

This is the mildest possible phase-transition scaffolding: one control parameter and one square-root order parameter.

6. Viable temperature range

The mild parameter scans prefer shallow temperature dependence, typically with exponents like $m = 1$ or $m = 1/2$, and place the viable transition regime in the neighborhood

$$T_c \sim 5000\text{–}10000\,\mathrm{K}.$$

This lies above but near the optical recombination temperature and gives relic phase amplitudes of order unity rather than pathological values.

7. 3D-to-2D projection onto the last-scattering surface

The relic phase perturbation $\delta\Phi(\mathbf{r})$ is a three-dimensional source field. The observed CMB anisotropy is a field on the two-dimensional last-scattering sphere:

$$\frac{\Delta T}{T}(\psi).$$

So the mapping cannot be pointwise. It must be a projection integral.

The simplest bounded source model is a spherical Gaussian event:

$$\delta\Phi(r) = A \exp\left[-\frac{r^2}{2\sigma^2}\right].$$

Its Laplacian is

$$\nabla^2\delta\Phi = \delta\Phi\left(\frac{r^2}{\sigma^4} - \frac{3}{\sigma^2}\right).$$

The simplest thin-shell proxy is then

$$\left.\frac{\Delta T}{T}\right|_{\text{proxy}}(\psi) = B\,\nabla^2\delta\Phi\big(r(\psi)\big),$$

where B is the transfer coefficient.

8. Raw projection: geometry survives, amplitude fails

With the benchmark choices

$$A = 1.2 \times 10^{-4}, \qquad \sigma/R_{\text{lss}} = 0.1, \qquad D/R_{\text{lss}} = 1, \qquad B = 1,$$

the raw projection gives a peak amplitude of approximately

$$\left|\frac{\Delta T}{T}\right|_{\text{peak}} \approx 0.036,$$

compared to the characteristic CMB scale

$$10^{-5}.$$

So the raw readout overshoots by roughly a factor of

$$3600.$$

At the same time, the angular width is about 5°, so the geometry survives while the amplitude fails catastrophically.

This rules out the naive identification

$$\frac{\Delta T}{T} = \nabla^2\delta\Phi$$

sampled sharply on a razor-thin shell with $B = 1$.

9. Filtered projection

The physically obvious suppressions are:

1. finite shell thickness,
2. angular beam smoothing,
3. sub-unity transfer efficiency.

So the filtered observable is written as

$$\frac{\Delta T}{T} = B \cdot \text{smoothed}\left[\text{shell-average}\left(\nabla^2 \delta\Phi\right)\right].$$

A sweep was performed over

$$\tau/R_{\text{lss}} \in \{0.005, 0.01, 0.02, 0.05\},$$

$$\theta_{\text{beam}} \in \{0.5^\circ, 1^\circ, 2^\circ, 5^\circ\},$$

$$B \in \{10^{-4}, 3\times10^{-4}, 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}, 10^{-1}\}.$$

This gives 112 combinations in total.

The outcome is that 56 of 112 combinations land in the target 0.1× to 10× CMB amplitude window, with best-fit regions clustered near

$$\tau/R_{\text{lss}} \sim 0.005\text{–}0.02, \qquad \theta_{\text{beam}} \sim 0.5^\circ\text{–}2^\circ, \qquad B \sim 3\times10^{-4}.$$

The angular widths remain around 5° to 5.25°.

So the filtered projection says:

- the 3D-to-2D idea survives,
- the raw observable was too strong,
- the target amplitude is reached with a stable transfer coefficient near 3×10^{-4}.

10. Transfer-budget extraction

A direct transfer-budget extraction compares the raw and filtered projections. The geometric/detector suppression in the best rows remains close to unity, typically

$$0.92\text{–}0.99,$$

and only drops to about

$$0.68$$

in the strongest smoothing cases.

So the needed coefficient

$$B_{\text{therm}} \sim 3 \times 10^{-4}$$

is not mostly geometry or detector blur. It behaves like a real thermodynamic conversion factor.

11. Compound transfer coefficient

The surviving working form is

$$B_{\text{therm}} = q_T f_b f_r \varepsilon,$$

where:

- $q_T = 1/4$ is the radiation temperature-conversion factor from $\rho_\gamma \propto T^4$,
- f_b is the matter/baryonic fraction relevant to the shatter,
- f_r is the radiation fraction at that epoch,
- ε is the residual transfer efficiency.

12. Partition tests H1–H4

The partition tests are:

$$\begin{aligned}
&\text{H1: } B = f_b, \\
&\text{H2: } B = f_b f_r, \\
&\text{H3: } B = q_T f_b f_r, \\
&\text{H4: } B = q_T f_b f_r \varepsilon.
\end{aligned}$$

The scan results are:

- H1: zero hits,
- H2: zero hits,
- H3: a few grazing hits,
- H4: 50 hits in the target ballpark.

So simple abundance partition is ruled out, and the compound architecture survives.

13. The candidate $1/(4\pi^2)$ splice

The numerically preferred residual efficiency is

$$\varepsilon \approx 0.03.$$

This is suggestively close to

$$\frac{1}{4\pi^2} = 0.02533.$$

So the candidate identification is

$$\varepsilon_{\text{candidate}} = \frac{1}{4\pi^2},$$

which gives the candidate transfer law

$$B_{\text{therm,candidate}} = \frac{q_T f_b f_r}{4\pi^2}.$$

This splice produces 17 direct ballpark hits in the mild parameter ranges tested. Example points include

$$(f_b, f_r) = (0.15, 0.3) \Rightarrow B_{\text{pred}} = 2.85 \times 10^{-4},$$

$$(f_b, f_r) = (0.10, 0.5) \Rightarrow B_{\text{pred}} = 3.17 \times 10^{-4},$$

$$(f_b, f_r) = (0.20, 0.2) \Rightarrow B_{\text{pred}} = 2.53 \times 10^{-4}.$$

So the candidate splice is numerically viable and much cleaner than a purely phenomenological $\varepsilon \approx 0.03$ fit.

14. Angular-to-linear phase-space interpretation

The proposed interpretation of

$$\frac{1}{4\pi^2}$$

is that it encodes an angular-to-linear phase-space conversion.

The internal Compton clock is angular, with one full cycle spanning

$$2\pi.$$

The observed CMB anisotropy is a linear thermodynamic observable in the bulk projected onto a two-dimensional last-scattering surface.

The candidate interpretation is therefore that converting an internal angular variance into an

external linear temperature observable requires dividing by the product of the two angular domains:

$$(2\pi)^2 = 4\pi^2.$$

This is the physical motivation of the splice.

15. Exact epistemic status

The early-universe sector has reached the following status:

- the numerical match is strong,
- the physical interpretation is motivated,
- the splice is cleaner than simpler alternatives,
- but the unbroken symbolic derivation is not yet in hand.

The missing symbolic chain is

$$\nabla^2 \delta\Phi \longrightarrow \text{photon-fluid response} \longrightarrow \frac{\Delta T}{T},$$

with the $1/(4\pi^2)$ factor attached to an explicit angular-to-linear conversion step.

So the correct statement is:

$$\boxed{\text{The } \frac{1}{4\pi^2} \text{ splice is stronger than a coincidence, weaker than a theorem.}}$$

This is exactly why V10 keeps it at candidate-derivation status and names it as Open Theorem 3 rather than as a closed result.

16. What this chapter has established

This chapter has established the following:

1. The leading bridge between the cosmological screening sector and the relic phase field is the identity-like bridge

$$\delta\chi = \delta\Phi, \qquad \bar{\Phi} = \ln a, \qquad \dot{\bar{\Phi}} = H, \qquad U_{\text{screen}} \approx \alpha\chi.$$

2. This bridge is quantitatively constrained by recombination-era amplitudes and does not require pathological coefficients.

3. The shatter admits a mild Landau-style phase-transition scaffolding with viable temperatures in the 5000–10000 K neighborhood.

4. The raw 3D-to-2D projection overshoots the CMB amplitude badly, but the filtered projection survives and lands in the right ballpark for a stable transfer coefficient near 3×10^{-4}.

5. Transfer-budget extraction shows that the needed coefficient is not mostly a detector artifact.

6. Partition tests rule out simple abundance ratios and favor the compound architecture

$$B_{\text{therm}} = q_T f_b f_r \varepsilon.$$

7. The candidate splice

$$B_{\text{therm,candidate}} = \frac{q_T f_b f_r}{4\pi^2}$$

is numerically viable, physically motivated, and not yet a closed theorem.

So the early-universe sector should currently be read as:

- structurally parallel to the late-universe engine crash,
- quantitatively constrained by mildness and projection tests,
- and carrying a numerically serious but still not fully closed fifth appearance of $4\pi^2$.

Companion Derivations to Solvency Field Theory (V10)

C12. Strong-Field Closure and the Schwarzschild/TOV Recovery

Master section mirrored: Section 12
Claim being derived: In the strong-field regime, the local information-maintenance balance closes onto the standard static spherically symmetric relativistic geometry. The resulting exterior metric is exactly Schwarzschild, and the interior hydrostatic equilibrium law is exactly Tolman-Oppenheimer-Volkoff.

Inputs / assumptions:

1. Static spherical symmetry.
2. Asymptotic flatness.
3. Local maintenance-cost additivity shell-by-shell.
4. Multiplicative redshift composition for successive radial layers.
5. Perfect-fluid matter source in the interior.
6. Strong-field closure interpreted as the regime in which local screening is complete and the local balance law dominates over cosmological screening corrections.

Result: The strong-field solvent metric takes the exponential form

$$g_{tt} = -e^{-2U(r)}, \qquad g_{rr} = e^{+2U(r)}$$

in the simplest reciprocal gauge, or equivalently the standard Schwarzschild-form gauge

$$ds^2 = -\Big(1 - \frac{2Gm(r)}{c^2 r}\Big)e^{2\Phi(r)}c^2dt^2 + \Big(1 - \frac{2Gm(r)}{c^2 r}\Big)^{-1}dr^2 + r^2 d\Omega^2.$$

In exterior vacuum,

$$m(r) = M, \qquad e^{2\Phi(r)} = 1,$$

so the metric reduces exactly to

$$ds^2 = -\Big(1 - \frac{2GM}{c^2 r}\Big)c^2dt^2 + \Big(1 - \frac{2GM}{c^2 r}\Big)^{-1}dr^2 + r^2 d\Omega^2.$$

In the interior, covariant conservation gives

$$\frac{dp}{dr} = -(\rho c^2 + p)\frac{d\Phi}{dr},$$

and with the standard mass function

$$\frac{dm}{dr} = 4\pi r^2 \rho,$$

the hydrostatic balance law becomes the exact TOV equation

$$\frac{dp}{dr} = -\frac{G\left(\rho + \frac{p}{c^2}\right)\left(m + \frac{4\pi r^3 p}{c^2}\right)}{r\left(r - \frac{2Gm}{c^2}\right)}.$$

Status: Leading strong-field closure of the current SFT scaffold.

C12.1 Local compactness as the regime variable

The organizing variable for the strong-field sector is the local compactness

$$\chi(r) = \frac{Gm(r)}{2c^2 r}.$$

Its role is to mark the transition between the weak-field and strong-field sectors:

- $\chi \ll 1$: weak-field screening regime,
- $\chi \to \frac{1}{2}$: Schwarzschild/horizon regime,
- $0 < \chi < \frac{1}{2}$: continuous compact-object regime from planets through neutron stars.

The role of χ in the companion is not merely classificatory. It is the natural dimensionless source variable for any local solvent potential, because it measures the ratio of gravitational binding scale to rest-energy scale. In the weak field it reduces to the familiar Newtonian potential scale up to an order-unity normalization, while in the strong field it approaches the horizon limit.

C12.2 Local balance and the solvent potential

The cosmological master equation

$$H = \frac{\dot{S}}{I}$$

is a global bookkeeping relation. In the strong-field regime, the local analogue is that at each radius, the local maintenance production must be absorbable by the local geometry, otherwise the ledger is insolvent at that radius.

For the companion, the cleanest way to encode that local closure is through a scalar solvent potential $U(r)$, defined so that:

- U adds shell-by-shell,
- local redshift or maintenance attenuation compounds multiplicatively across shells.

If a thin radial shell contributes an incremental maintenance burden dU, then the redshift factor across that shell multiplies by

$$e^{-2\,dU}.$$

Successive shells therefore compose as

$$f(r + dr) = f(r)e^{-2\,dU},$$

so

$$d\ln f = -2\,dU.$$

Integrating gives

$$\ln f(r) = -2U(r) + \text{const}.$$

Asymptotic flatness requires $f(\infty) = 1$, so the integration constant vanishes, and therefore

$$\boxed{f(r) = e^{-2U(r)}.}$$

This is the uniqueness pivot for the exponential form. It does not come from guessing an exponential and hoping it works; it comes from the fact that the local cost adds, the metric attenuation multiplies, and the only function that turns additive source increments into multiplicative response is the exponential.

C12.3 Static spherical metric ansatz

The minimal reciprocal static spherical gauge is therefore

$$ds^2 = -e^{-2U(r)}c^2dt^2 + e^{+2U(r)}dr^2 + r^2d\Omega^2.$$

This form is useful because it makes the exponential closure transparent:

$$g_{tt}g_{rr} = -1.$$

But for comparison with standard relativistic stellar structure, it is more convenient to pass to the usual mass-function gauge

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \left(1 - \frac{2Gm(r)}{c^2r}\right)^{-1}dr^2 + r^2d\Omega^2.$$

These are not different physical sectors. They are two standard ways of parameterizing the same static spherical closure.

C12.4 Exterior vacuum closure

Outside a compact spherical body,

$$\rho(r) = 0, \qquad p(r) = 0, \qquad m(r) = M = \text{const.}$$

The exterior solvent geometry must satisfy three conditions:

1. **asymptotic flatness**
$$g_{tt} \to -1, \qquad g_{rr} \to 1 \quad \text{as} \quad r \to \infty;$$

2. **Newtonian matching**
$$g_{tt} \approx -(1 + 2\phi_N/c^2) \quad \text{for} \quad \frac{GM}{rc^2} \ll 1,$$
with
$$\phi_N = -\frac{GM}{r};$$

3. **single-parameter compactness sourcing** - the vacuum exterior depends only on the enclosed mass M.

The unique standard static spherical vacuum form satisfying these conditions is

$$\boxed{ds^2 = -\Big(1 - \frac{2GM}{c^2 r}\Big)c^2 dt^2 + \Big(1 - \frac{2GM}{c^2 r}\Big)^{-1} dr^2 + r^2 d\Omega^2.}$$

This is exact Schwarzschild.

To see the weak-field matching explicitly, expand

$$1 - \frac{2GM}{c^2 r} = 1 + \frac{2\phi_N}{c^2}.$$

Since $\phi_N = -GM/r$, the standard Newtonian redshift limit is recovered immediately.

In the exponential-potential language, this corresponds to

$$e^{-2U(r)} = 1 - \frac{2GM}{c^2 r},$$

so

$$U(r) = -\frac{1}{2}\ln\Big(1 - \frac{2GM}{c^2 r}\Big).$$

Thus the exponential closure and the exact Schwarzschild exterior are perfectly compatible: the strong-field closure does not compete with Schwarzschild, it lands on it.

C12.5 Why the exponential closure is the right strong-field statement

The deeper statement is:

- in the weak field, one often works with additive potentials;
- in the strong field, the physically relevant quantity is not the additive potential itself but the multiplicative redshift/attenuation accumulated across shells.

If the local Landauer cost is absorbed shell-by-shell, and if each shell contributes independently to the total redshift factor, then the shell composition law is multiplicative. That is what forces exponentiation.

So the strong-field closure is best understood as the local analogue of the global SFT logic:

- global bulk processing is compared to global boundary accommodation;
- local shell maintenance is compared to local curvature accommodation;
- additive costs become multiplicative metric response.

That is the structural reason the strong-field solution closes onto standard GR rather than deviating from it.

C12.6 Interior perfect-fluid configuration

Now consider a static spherically symmetric perfect fluid with density $\rho(r)$ and pressure $p(r)$. Write the metric in standard form:

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \left(1 - \frac{2Gm(r)}{c^2r}\right)^{-1}dr^2 + r^2d\Omega^2.$$

The enclosed mass function is

$$\boxed{\frac{dm}{dr} = 4\pi r^2\rho.}$$

This is simply the definition of the Misner-Sharp mass in static spherical symmetry.

The hydrostatic balance law comes from covariant conservation of the perfect-fluid stress tensor,

$$\nabla_\mu T^{\mu\nu} = 0.$$

The radial component gives

$$\boxed{\frac{dp}{dr} = -(\rho c^2 + p)\frac{d\Phi}{dr}.}$$

This equation is purely kinematic/conservation-based: pressure gradients balance the redshift gradient encoded in $\Phi(r)$.

C12.7 From local solvent redshift to exact TOV

To close the system, one needs $d\Phi/dr$. In the present strong-field closure, the solvent geometry is identified with the standard static spherical form, so the redshift gradient is

$$\boxed{\frac{d\Phi}{dr} = \frac{Gm(r) + 4\pi Gr^3p/c^2}{c^2r^2\left(1 - \frac{2Gm(r)}{c^2r}\right)}.}$$

Substitute this into

$$\frac{dp}{dr} = -(\rho c^2 + p)\frac{d\Phi}{dr}.$$

Then

$$\frac{dp}{dr} = -(\rho c^2 + p)\frac{Gm(r) + 4\pi Gr^3p/c^2}{c^2r^2\left(1 - \frac{2Gm(r)}{c^2r}\right)}.$$

Divide numerator and denominator by c^2 in the first factor:

$$\rho c^2 + p = c^2\left(\rho + \frac{p}{c^2}\right).$$

So

$$\frac{dp}{dr} = -\frac{G\left(\rho + \frac{p}{c^2}\right)\left(m(r) + \frac{4\pi r^3 p}{c^2}\right)}{r^2\left(1 - \frac{2Gm(r)}{c^2 r}\right)}.$$

Using

$$r^2\left(1 - \frac{2Gm}{c^2 r}\right) = r\left(r - \frac{2Gm}{c^2}\right),$$

this becomes the standard TOV equation:

$$\boxed{\frac{dp}{dr} = -\frac{G\left(\rho + \frac{p}{c^2}\right)\left(m + \frac{4\pi r^3 p}{c^2}\right)}{r\left(r - \frac{2Gm}{c^2}\right)}.}$$

Thus the strong-field closure recovers the exact relativistic hydrostatic equilibrium law.

C12.8 Exterior/interior matching

At the stellar surface $r = R_\star$, where

$$p(R_\star) = 0,$$

the interior solution must match continuously onto the exterior vacuum Schwarzschild solution with total mass

$$M = m(R_\star).$$

So the full compact-object closure is:

- solve

$$\frac{dm}{dr} = 4\pi r^2 \rho, \qquad \frac{dp}{dr} = -\frac{G\left(\rho + \frac{p}{c^2}\right)\left(m + \frac{4\pi r^3 p}{c^2}\right)}{r\left(r - \frac{2Gm}{c^2}\right)}$$

  inside the fluid,

- then match to

$$ds^2 = -\Big(1 - \frac{2GM}{c^2 r}\Big)c^2 dt^2 + \Big(1 - \frac{2GM}{c^2 r}\Big)^{-1} dr^2 + r^2 d\Omega^2$$

  outside.

This is exactly the standard relativistic compact-object problem.

C12.9 Relation back to Branch C*

The strong-field closure is not isolated from the weak-field lag-action program. The clean way to state the relation is:

- **Branch C*** is the weak-field lag-action sector that reproduces the standard first post-Newtonian gates,
- **C12** is the strong-field limit in which local screening is complete and the solvent geometry becomes exact Schwarzschild/TOV.

So the framework picture is not "weak-field theory here, unrelated strong-field theory there." It is one screening/maintenance principle with two regimes:

- incomplete local absorption in the weak field,
- complete local absorption in the strong field.

C12.10 What is proved and what is assumed

This chapter establishes the following clearly:

1. If the local maintenance burden adds shell-by-shell and the corresponding metric attenuation composes multiplicatively, then the solvent metric response must exponentiate:
$$f(r) = e^{-2U(r)}.$$
2. In exterior spherical vacuum, the asymptotically flat strong-field closure is exact Schwarzschild:
$$ds^2 = -\Big(1 - \frac{2GM}{c^2 r}\Big)c^2 dt^2 + \Big(1 - \frac{2GM}{c^2 r}\Big)^{-1} dr^2 + r^2 d\Omega^2.$$
3. For a perfect-fluid interior, covariant conservation plus the same closure yields exact TOV:
$$\frac{dp}{dr} = -\frac{G\left(\rho + \frac{p}{c^2}\right)\left(m + \frac{4\pi r^3 p}{c^2}\right)}{r\left(r - \frac{2Gm}{c^2}\right)}.$$

What is still assumed in the current scaffold is the precise local-balance functional that produces the static spherical redshift gradient in full generality. That is why V10 correctly calls this the **leading strong-field closure** and places the full symbolic uniqueness proof in the companion rather than pretending the main paper already displayed it line by line.

C12.11 What this chapter has established

The strong-field sector of SFT closes onto standard general relativity rather than modifying it.

The logical chain is:

- local maintenance costs add shell-by-shell,
- redshift response composes multiplicatively,

- multiplicative composition forces exponential metric response,
- the asymptotically flat spherical vacuum closure is exact Schwarzschild,
- the static perfect-fluid interior closure is exact TOV.

So in the strong-field regime, SFT does not compete with GR. It supplies the information-maintenance foundation under which exact GR appears as the solvent local closure.

This exponent is important enough to state explicitly:

$$\boxed{\ell_P^2 \approx 2.61 \times 10^{-70}\ \mathrm{m}^2.}$$

Any appearance of $10^{-60}\,\mathrm{m}^2$ in development drafts is a typographical error and should be treated as superseded.

A.3 A.3 Horizon geometry

For flat FLRW minisuperspace,

$$R_H = \frac{c}{H},$$

$$A_H = 4\pi R_H^2 = 4\pi \left(\frac{c}{H}\right)^2,$$

$$V_H = \frac{4\pi}{3} R_H^3 = \frac{4\pi}{3}\left(\frac{c}{H}\right)^3.$$

The associated holographic information capacity is

$$I = \frac{A_H}{4\ell_P^2} = \frac{\pi c^2}{\ell_P^2 H^2}.$$

The companion uses the Hubble sphere as the apparent-horizon bookkeeping boundary of the FLRW reduction, consistent with the master's use of one-Hubble-time settlement in the count-equation formulation.

A.4 A.4 Compton maintenance rate

For total enclosed mass-energy M,

$$\dot{S} = \frac{Mc^2}{h}.$$

For Hubble-volume matter with mean density ρ,

$$M = \rho V_H, \qquad \dot{S} = \frac{\rho V_H c^2}{h}.$$

This counts complete Compton cycles per unit time, using the linear-cycle convention.

A.5 A.5 Modified Friedmann backbone

The core modified Friedmann equation derived in C5 is

$$H^2 = \frac{2G\rho}{3\pi}.$$

Compared with the standard flat-dust Friedmann equation

$$H_{\mathrm{std}}^2 = \frac{8\pi G\rho}{3},$$

the ratio is

$$\frac{H_{\mathrm{SFT}}^2}{H_{\mathrm{std}}^2} = \frac{1}{4\pi^2}.$$

This is Appearance 1 of the structural identity and the backbone of later weak-field and cosmological-sector results.

A.6 A.6 Weak-field threshold

The weak-field acceleration scale used in the SPARC sector is

$$a_0 \sim \frac{cH_0}{2\pi}.$$

This exposes one factor of 2π because the acceleration threshold inherits the square root of the background $1/(4\pi^2)$ ratio.

A.7 A.7 Cosmological reconstruction formulas

The corrected framework-level cosmological reconstruction uses

$$H(z) = \frac{\dot{S}_{\mathrm{total}}(z)}{I(z)}, \qquad E(z)^2 = \frac{H(z)^2}{H_0^2}.$$

The total effective equation of state is

$$w_{\mathrm{eff}}(z) = -1 + \frac{1+z}{3}\frac{d\ln H^2}{dz}.$$

The dark-energy residual is

$$\Omega_{\mathrm{DE}}(z) = E(z)^2 - \Omega_{m0}(1+z)^3,$$

and the corresponding reconstructed dark-energy equation of state is

$$w_{\mathrm{DE}}(z) = -1 + \frac{1+z}{3\,\Omega_{\mathrm{DE}}(z)}\frac{d\Omega_{\mathrm{DE}}}{dz}.$$

These formulas are the mathematically corrected chain now carried alongside the historical pipeline, with full numerical reconciliation still explicitly under audit in the V10 master.

A.8 A.8 Strong-field formulas

For static spherical symmetry, the standard mass-function gauge is

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \left(1 - \frac{2Gm(r)}{c^2r}\right)^{-1} dr^2 + r^2d\Omega^2.$$

The enclosed mass function satisfies

$$\frac{dm}{dr} = 4\pi r^2\rho,$$

and hydrostatic equilibrium is

$$\frac{dp}{dr} = -(\rho c^2 + p)\frac{d\Phi}{dr}.$$

The resulting exact TOV equation is

$$\frac{dp}{dr} = -\frac{G\left(\rho + \frac{p}{c^2}\right)\left(m + \frac{4\pi r^3 p}{c^2}\right)}{r\left(r - \frac{2Gm}{c^2}\right)}.$$

These formulas are part of the leading strong-field closure that V10 explicitly presents as the GR-recovering capstone of the current scaffold.

A.9 A.9 Unit checks that recur often

The following dimensional identities recur throughout the companion:

$$[G\rho] = T^{-2},$$

$$\left[\frac{Mc^2}{h}\right] = T^{-1},$$

$$\left[\frac{A_H}{4\ell_P^2}\right] = 1,$$

$$\left[\frac{E_H}{hH}\right] = 1.$$

These are worth listing because they make later dimensional checks immediate.

B. Appendix B. Script and Data Inventory

Purpose: Provide a compact audit map from each major derivation or numerical result in the companion to the supporting scripts, data products, and expected outputs.

This appendix is deliberately split into two categories:

- **current preserved scripts** created during the V10 audit and companion build,
- **historical pipeline components** whose filenames may vary across local storage but whose role in the framework is fixed.

B.1 B.1 Current preserved scripts

B.1.1 B.1.1 Early-universe / CMB sector

1. `sft_3d_to_2d_filtered_projection_sweep.py`
   Supports: C11.7
   Purpose: finite-shell + beam-smoothed projection sweep for relic phase events
   Primary outputs: filtered amplitude table, best-profile plot, metadata

2. `sft_transfer_budget_extraction.py`
   Supports: C11.8
   Purpose: separates raw-vs-filtered amplitude suppression into geometric suppression and residual transfer factor
   Primary outputs: transfer-budget summary and metadata

3. `sft_Btherm_partition_test.py`
   Supports: C11.9–C11.10
   Purpose: tests partition hypotheses $H1$–$H4$ for the thermodynamic transfer coefficient
   Primary outputs: partition-test summary and metadata

4. `sft_goal3_transfer_splice_test.py`
   Supports: C11.11
   Purpose: direct splice test for
   $$B_{\text{therm}} = \frac{q_T f_b f_r}{4\pi^2}$$
   Primary outputs: splice-test summary and metadata

B.1.2 B.1.2 Cosmological corrected-chain audit

5. `sft_corrected_wde_reconstruction.py`
   Supports: C10.12–C10.14
   Purpose: reconstructs
   $$H(z) \to \Omega_{\text{DE}}(z) \to w_{\text{DE}}(z)$$
   from channel-history input and fits the resulting CPL pair
   Primary outputs: corrected reconstruction summary, reconstructed $w_{\text{DE}}(z)$ table, input audit table, metadata

B.2 B.2 Historical pipeline components

The following items are part of the historical numerical backbone of the framework but may not yet be packaged under standardized filenames. They should still be listed in the repro bundle because they support claims carried by V10.

B.2.1 B.2.1 Historical TNG300 channel extraction

Supports: C10.6–C10.11
Role: active/passive channel histories from TNG300 snapshots

Required data products: snapshot redshifts, star-formation history, black-hole accretion history, M_{200c} history

B.2.2 B.2.2 Historical DESI-facing CPL fit pipeline

Supports: C10.11
Role: produces the historical

$$(w_0, w_a) = (-0.837, -0.615)$$

result and associated crossing redshift

B.2.3 B.2.3 Structural-specificity comparison

Supports: C10.9
Role: comparison of M_{200c} versus M_{halo} in the passive channel

B.2.4 B.2.4 Reservoir-ablation comparison

Supports: C10.10
Role: shows catastrophic degradation when the passive channel is forced static rather than dynamical

B.2.5 B.2.5 Weak-field / SPARC scripts

Supports: C6
Role: baryonic-only vs corrected rotation-curve comparison, interpolation function application, λ-morphology audit

B.3 B.3 Inventory template for final repro bundle

When the local repro bundle is assembled, every script entry should have the following fields:

Script / notebook	Companion chapter	Input data	Output files	Status
filename	Cx.y	file(s) used	file(s) produced	preserved / historical / under audit

That template is more important than having perfect filenames immediately. It makes the bundle auditable even if the original local script organization was messy.

B.4 B.4 Data dependencies

The minimum data dependencies implied by the current companion are:

- **SPARC** rotation-curve sample
- **IllustrisTNG TNG300** snapshot histories
- **Planck 2018** CMB comparison products
- **DESI DR2** CPL region / likelihood comparison products

These are not new theoretical ingredients; they are the empirical references the framework is already using in the master.

C. Appendix C. Superseded and Rejected Derivation Paths

Purpose: Preserve development history without letting superseded paths contaminate the main derivation chapters.

This appendix is not a graveyard for every idea ever tried. It records the specific paths whose failure or supersession materially improved the framework.

C.1 C.1 Superseded instantaneous rate-law formulation

An earlier cosmological attempt wrote SFT as an instantaneous rate law, schematically

$$\frac{d}{dt}\left(\frac{A_H}{4\ell_P^2}\right) = \gamma\,(\text{bulk rate}),$$

with an unexplained coefficient γ that had to be hand-set to recover the correct modified Friedmann coefficient. In practice $\gamma = 3$ was needed.

This path is superseded because it was structurally incomplete: it tried to impose count balance instantaneously rather than over the one-Hubble-time settlement interval. The holographic count equation resolved the problem by making the causal interval explicit, at which point the missing factor emerged geometrically from the ratio between the Hubble sphere's volume and surface area. V10 explicitly preserves this supersession story as part of the count-equation motivation.

Reason for supersession: missing causal settlement interval

C.2 C.2 Older direct $w(t)$ mapping

Earlier development work used a direct working formula attempting to map $\dot{S}$, I, and ρ immediately to an effective $w(t)$. That historical map supported the original cosmology pipeline and helped generate the first DESI-facing fit, but the later audit showed that the mathematically correct framework-level reconstruction must proceed through the expansion history $H(z)$ and its derivatives.

So the direct older $w(t)$ formula is preserved as part of the historical pipeline, but is superseded as the preferred formal derivation. This is exactly why V10 now carries both:

- the historical pipeline,
- and the corrected $H(z) \to w_{\mathrm{DE}}(z)$ chain,

with full reconciliation flagged as a V10.1 audit item.

Reason for supersession: formal reconstruction must depend on the evolution of $H(z)$, not just on instantaneous $\dot{S}/I$

C.3 C.3 Coarse cosmological readout before the four-state ladder

Before the current four-state ladder, the cosmological sector used a coarser structural decomposition. That earlier bimodal or coarse-grained picture was useful in development, but V10 explicitly says the four-state ladder is now the controlling structural closure of the cosmological sector, and the bimodal story survives only as historical background.

Reason for supersession: the four-state ladder carries more structure, stronger physical specificity, and cleaner links to the projection-integral mechanism

C.4 C.4 Rejected simple transfer-coefficient hypotheses

The early-universe sector tested simpler possibilities for the CMB transfer coefficient:

$$B = f_b, \qquad B = f_b f_r, \qquad B = q_T f_b f_r.$$

The first two fail outright in the relevant target range, and the third only grazes the target. What survives is the compound architecture

$$B_{\mathrm{therm}} = q_T f_b f_r \varepsilon,$$

with the current candidate splice

$$\varepsilon_{\mathrm{candidate}} = \frac{1}{4\pi^2}.$$

So the companion preserves the rejected alternatives, but only as failed audit branches, not as living options.

Reason for rejection: quantitatively insufficient

C.5 C.5 Raw 3D→2D observable as naked Laplacian readout

A naive early-universe readout treated the observed CMB anisotropy as essentially the naked Laplacian of the relic phase field evaluated on a razor-thin shell. The geometry of the projected feature was interesting, but the amplitude overshot the target by roughly a factor of 3600.

This path was rejected as the literal observable. What survived was the filtered projection program:

- finite shell thickness,
- angular smoothing,

- and a real transfer coefficient.

Reason for rejection: geometry viable, amplitude catastrophically wrong

C.6 C.6 Over-strong claim versions rejected during drafting

The framework repeatedly faced a choice between "push the claim harder" and "state the honest epistemic register." V10 consistently chose the latter in several places:

- Appearance 5 remains a **candidate splice**, not a fifth fully derived appearance.
- The corrected w_{DE} chain is carried **alongside** the historical pipeline, not falsely substituted for it before full reconciliation.
- The strong-field section presents exact GR recovery as the **leading strong-field closure**, not as an independent alternative gravity theory.

This appendix records those "not chosen" stronger versions as rejected rhetorical paths, because the distinction matters for how the current framework should be read.

D. Appendix D. Status Ledger

Purpose: Give a one-page audit map of the framework's major claims.

Result	Companion chapter	Status	Notes
Master equation $H = \dot{S}/I$ as native scaffold identity	C3–4	scaffold identity	native to stated axioms, not assumption-free
Modified Friedmann equation $H^2 = 2G\rho/(3\pi)$	C5	derived	exact algebraic result
Friedmann ratio $1/(4\pi^2)$	C5 / C7	derived	Appearance 1
Weak-field threshold $a_0 \sim cH_0/(2\pi)$	C6	derived + empirically confirmed	SPARC sector
Morphology-dependent λ range	C6	empirical + partial derivation	full morphology-to-λ mapping still open
Holographic count equation	C8	derived	causal one-Hubble-time settlement form
Constrained minisuperspace action	C8	derived at FLRW level	full field-theoretic extension open
Four-state ladder	C10	structural derivation	leading current cosmological closure

Result	Companion chapter	Status	Notes
Static projection corrections $s_{\text{wall}} = 1/3,\ s_{\text{filament}} = 2/3$	C10	derived	second-moment spherical projections
Historical CPL pair $(-0.837, -0.615)$	C10	historical empirical output	established by historical channel pipeline
Corrected $H(z) \to w_{\text{DE}}(z)$ chain	C10	derived formula / under audit numerically	V10.1 reconciliation item
Structural-specificity test M_{200c} vs M_{halo}	C10	empirically load-bearing	favors organized virialized structure
Reservoir-ablation comparison	C10	empirically load-bearing	static reservoir rejected
Early-universe identity-like bridge	C11	candidate derivation	quantitatively bounded
Landau-style shatter scaffold	C11	candidate derivation	viable mildness corner
Raw 3D→2D projection geometry	C11	derived test / rejected literal observable	overshoots amplitude
Filtered projection program	C11	numerically viable	56/112 ballpark combinations
Transfer-budget extraction	C11	numerically derived	B_{therm} not mostly detector artifact
Compound transfer coefficient $q_T f_b f_r \varepsilon$	C11	numerically favored	simple partition rejected
Candidate splice $B_{\text{therm}} = q_T f_b f_r/(4\pi^2)$	C11 / C7	candidate splice	Open Theorem 3 boundary
Strong-field Schwarzschild recovery	C12	leading strong-field closure	exact GR exterior recovered
Strong-field TOV recovery	C12	leading strong-field closure	exact GR interior hydrostatics recovered

D.1 D.1 Open-theorem boundary

The master paper explicitly separates genuine open theorems from empirical research targets. The companion follows that split. The most important open-theorem boundaries relevant to the current companion are:

- **Theorem 1:** formal unification of the shatter and the engine crash

- **Theorem 2:** second 2π from first principles
- **Theorem 3:** closure of the CMB transfer derivation and the exact status of the fifth $4\pi^2$ appearance
- **Theorem 4:** morphology-to-λ derivation
- **Theorem 5:** full projection-integral unification across galactic and cosmological sectors

That is the correct stopping point of the present companion: enough is closed to audit the framework seriously, and the remaining open edges are named explicitly rather than hidden.